



Comparative Analysis of Spatial Transcriptomics Approaches for Cell-Cell Communication

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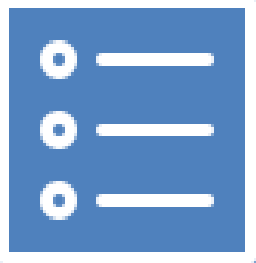
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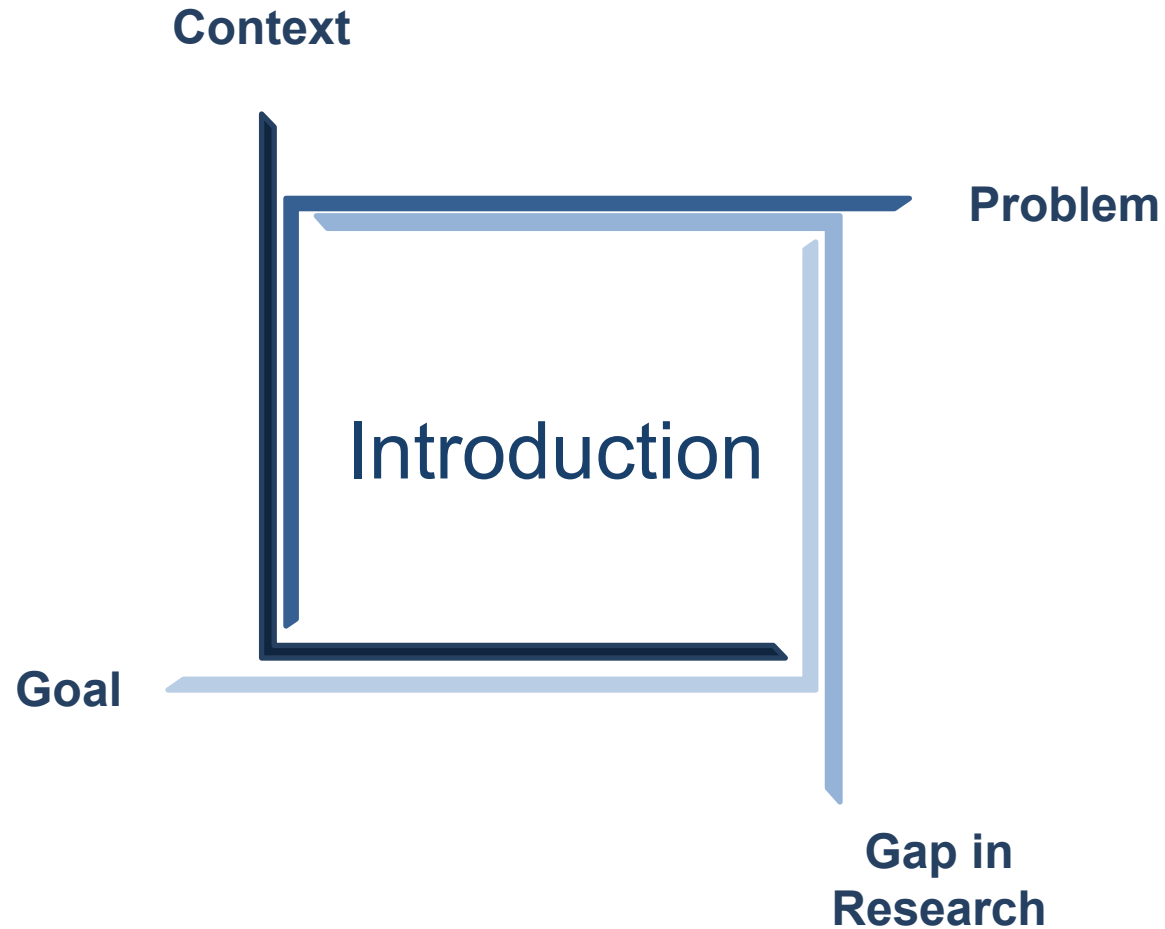
Presented by:

Akram Abushmais

Introduction	01
Background	02
Aim of work	03
Method	04
Results	05
Conclusion	06
Outcome	07
Future work	08



Outline



Spatial transcriptomics , Cell - Cell Communication , Ligand – Receptor signaling



Many CCC methods exist with different assumptions, no unified comparison

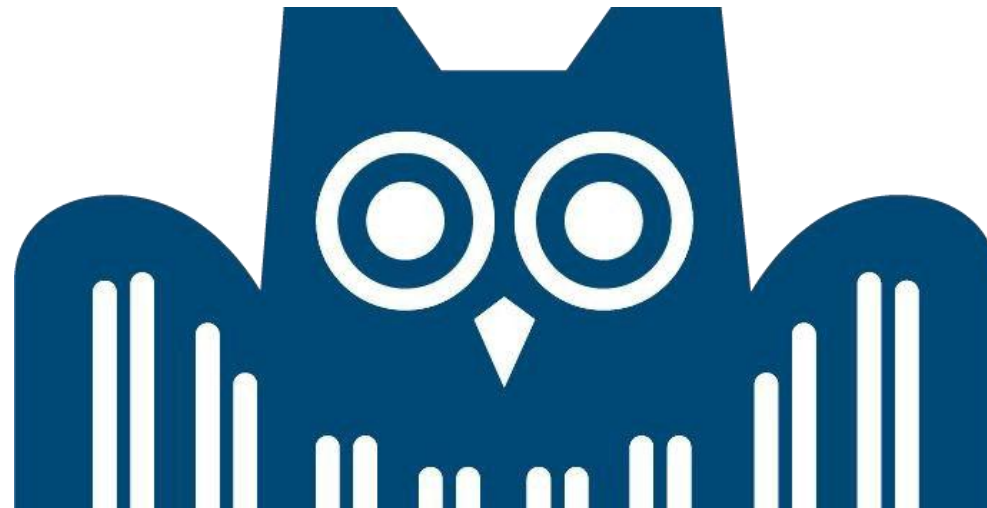


Existing benchmarks don't focus on spatial-native CCC tools



Compare spatial CCC methods on scalability, usability, consistency, & provide guidance

Background



Spatial transcriptomics

- Measures gene expression within tissue.
- Maintains the spatial location of cell or spot.
- Enables analysis of spatially organized biological processes.



Figure 1: Spatial transcriptomics

MERFISH hybridization

(Multiplexed Error-robust fluorescence)

It is method for imaging RNAs
using fluorescent probe that
hybridize to transcripts of interest
in fixed cells.

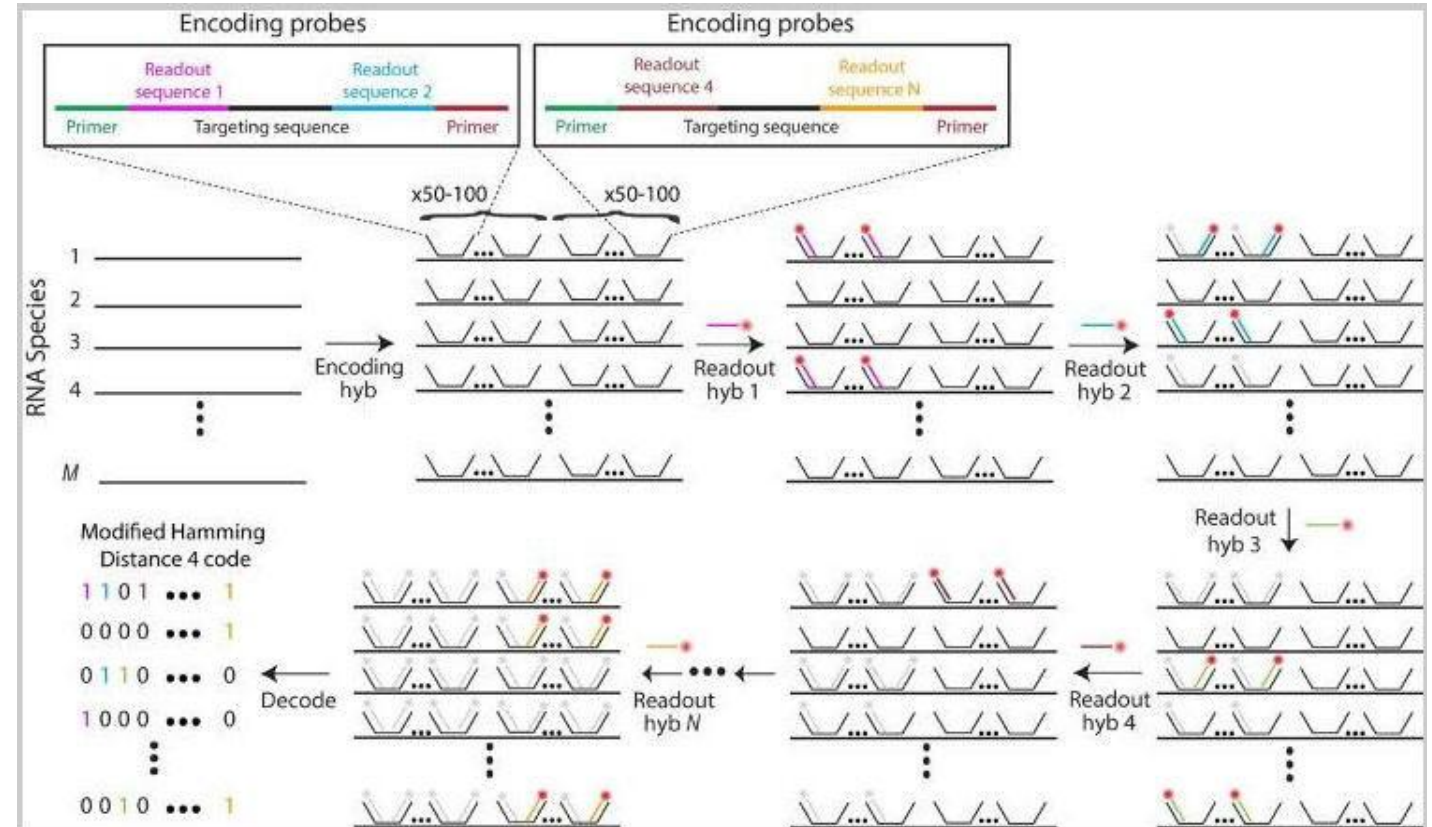


Figure 2: MERFISH technique

Sequence based

Stereo-seq captures RNA from tissue sections using DNA-nanoball barcoded arrays, enabling sequencing-based gene expression with ultra-high spatial resolution.

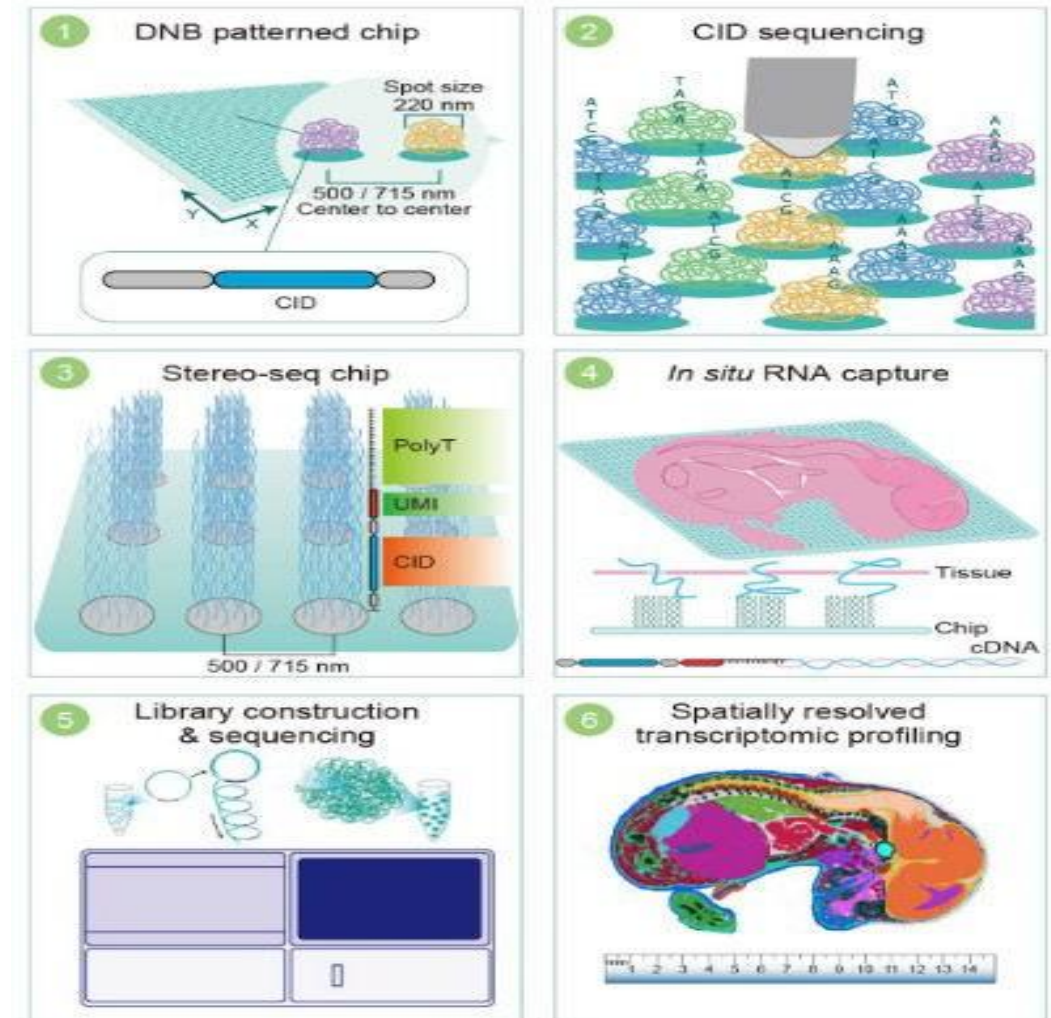
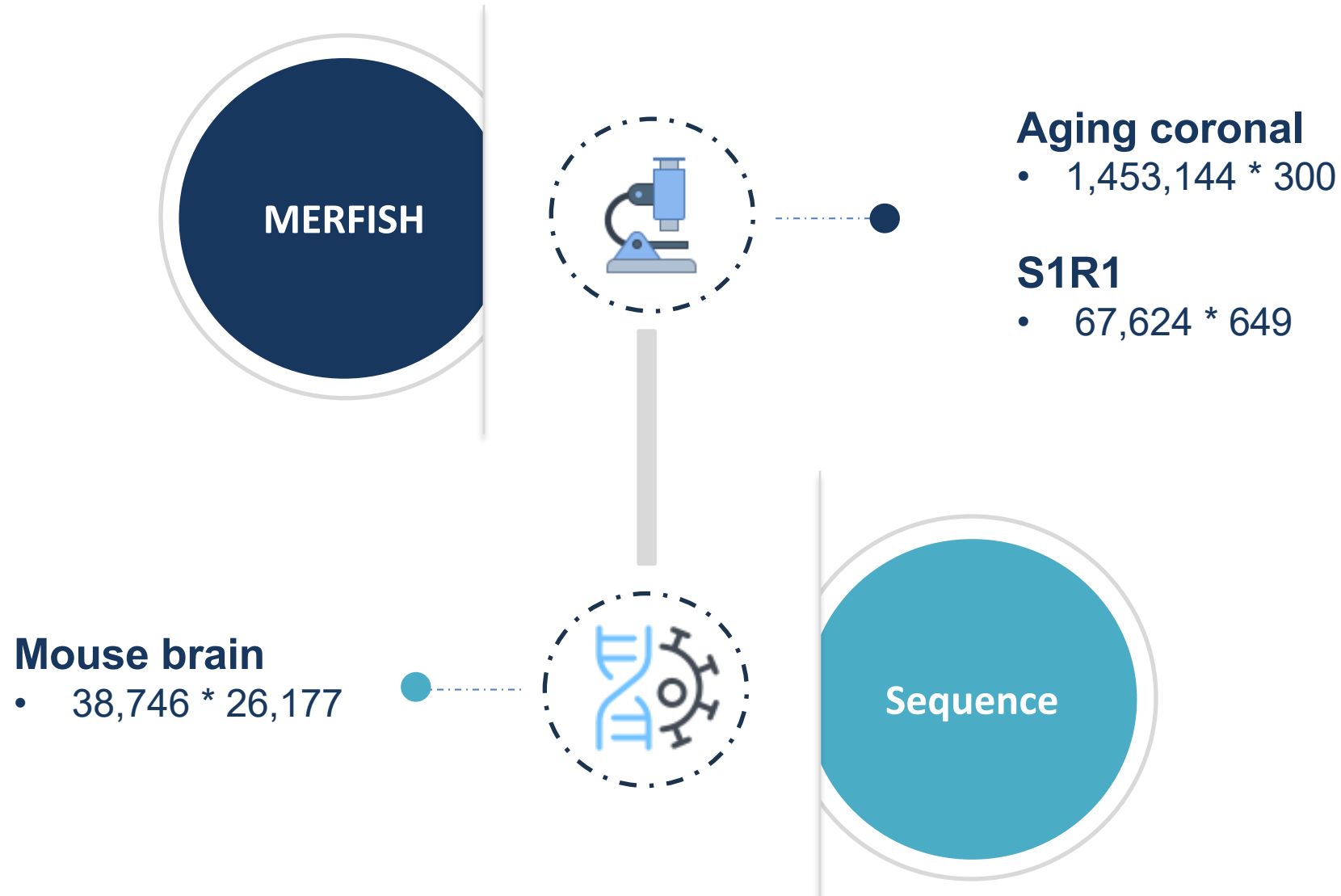


Figure 3: Stereo-seq pipeline

Data overview



Cell-Cell Communication

- Cells communicate via Ligand-receptor interaction.
- Communication can be direct or indirect.
- Spatial proximity is necessary for effective communication.

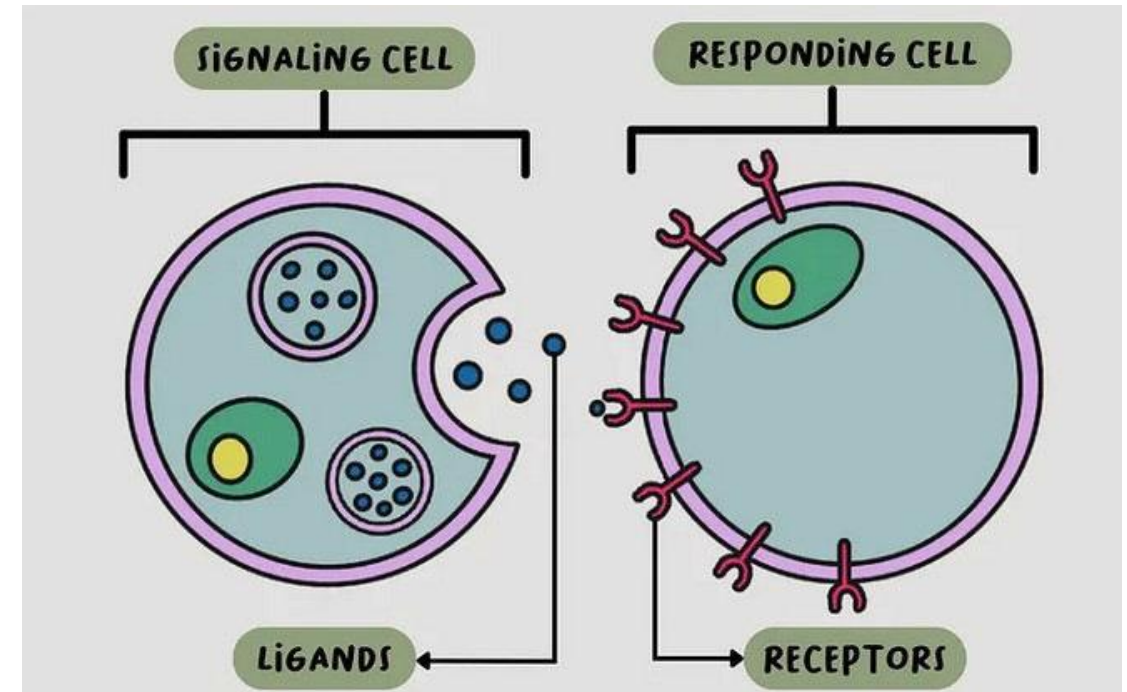


Figure 4: Components of Cellular Signaling

Evaluation of cell-cell interaction methods by integrating single-cell RNA sequencing data with spatial information

Zhaoyang Liu^{1 2}, Dongqing Sun^{1 2}, Chenfei Wang^{3 4}

Research

- Evaluates CCC tools by checking whether their predictions align with spatial proximity in ST data.
- Propose Distance Enrichment Score (DES) for evaluation.
- Focusing mainly on cluster level CCC prediction from scRNA tool.

Research gap

- Spatial transcriptomics is used for evaluation, not as a core modeling component.

ESICCC as a systematic computational framework for evaluation, selection, and integration of cell-cell communication inference methods

[Jiaxin Luo](#)^{1,2}, [Minghua Deng](#)³, [Xuegong Zhang](#)⁴, [Xiaoqiang Sun](#)^{2,✉}

Research

- Benchmark of CCC methods.
- Evaluate methods across scRNA and ST.
- Focus on LR prediction.

Research gap

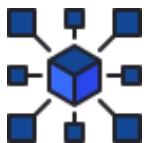
- Not focused exclusively on spatial CCC tools.
- Focus mainly on scRNA-seq based CCC methods.

What is the problem and how we solve it?

⚠ The Problems



Many CCC methods
Differ in assumptions



No unified framework
To guide researchers



Spatial-native tools
lack comparsion

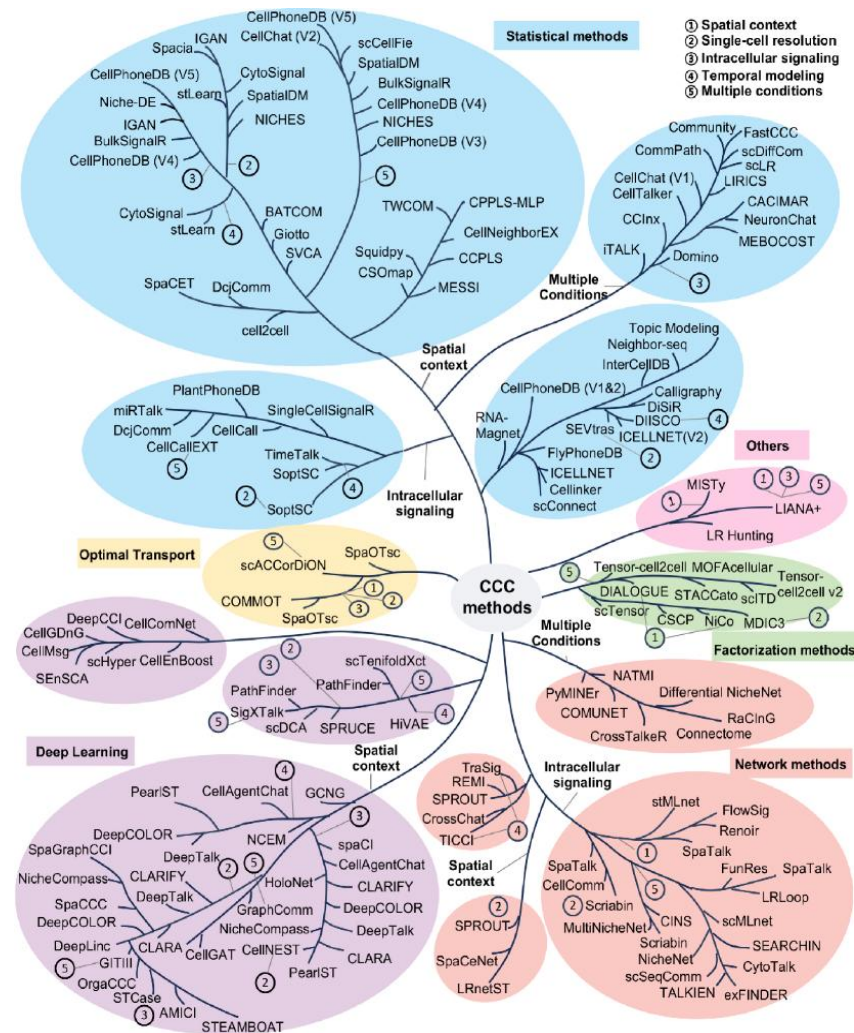


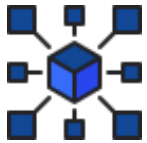
Figure 5: Spatial transcriptomics existing tools.

What is the problem and how we solve it?

⚠ The Problems



Many CCC methods
Differ in assumptions



No unified framework
To guide researchers



Spatial-native tools
lack comparison



Objectives

Compare native ST tools
COMMOT - spaCI - Spacia - VGAE-CCI

Unified evaluation pipeline
same data, same conditions

Guide method selection
by dataset size and research goal

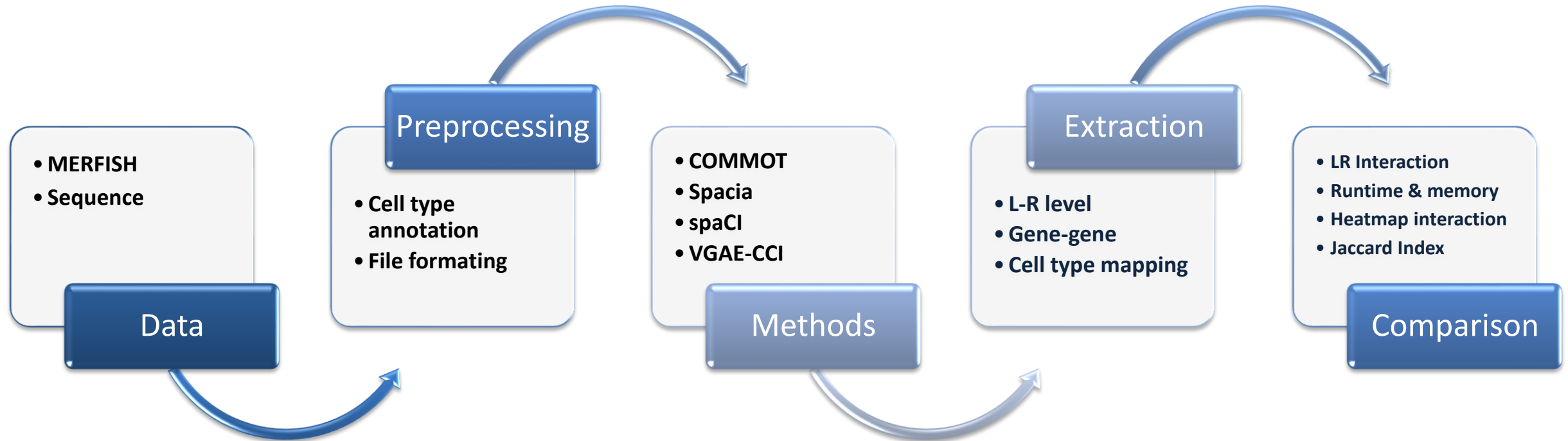


Thesis: Evaluated on unified pipeline on scalability, usability and consistency of inferred interaction

Methodology



Comparative Analysis of Spatial Transcriptomics Approach for Cell-Cell Communication Pipeline



COMMOT

(communication via Optimal transport)

- Treats ligand molecules as a mass to be transported from sender to receiver cells.
- Spatial distance between cells as transport cost.
- Combines gene expression and spatial distance.

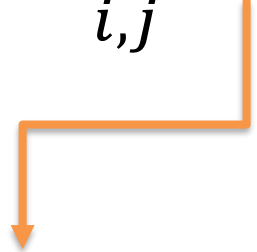
γ_{ij} : Total communication from cell $i \rightarrow j$

$P_{i,j,k,l}$: Amount of signal from cell i to j via ligand k to receptor l

C_{ij} : Spatial distance cost

K : Ligand gene

L : Receptor gene

$$\min_{\gamma} \sum_{i,j} \gamma_{ij} C_{ij}$$

$$\gamma_{ij} = \sum_{k,l} P_{i,j,k,l}$$

spaCI

(Graph attention-based cell-cell communication)

- Build spatial graph – cell are nodes, edge connect spatial neighbors.
- Combines gene expression embedding and spatial embedding to predict L-R interaction.

$$Z = \text{MLP}(Z_x \parallel Z_A)$$

Z_x : Gene expression embedding

Z_A : Spatial embedding

Z : Cell representation

MLP : Standard multilayer perceptron.

Spacia

(Bayesian multiple instance learning)

- The receiver cell is a bag and surrounding sender cells are instance.
- Learns which sender cells, and which of the expressed genes, drive expression change in the receiver.
- Uses MCMC to sample the full posterior distribution.

$$y_r = f \left(\sum_j w_j \cdot x_j \right) + \epsilon$$

y_r : Gene expression of receiver cell

x_j : Gene expression of sender cell

w_j : Signaling weight of sender j

f : Non-linear activation function

ε : Noise term.

VGAE-CCI

(Variational Graph Autoencoder)

- Encodes each cell into a low-dimensional latent vector using both gene expression and the spatial neighborhood graph.
- Cell-cell interaction is inferred from similarity between latent embeddings.

$$P(A_{ij} \mid z_i, z_j) = \sigma(z_i^T z_j)$$

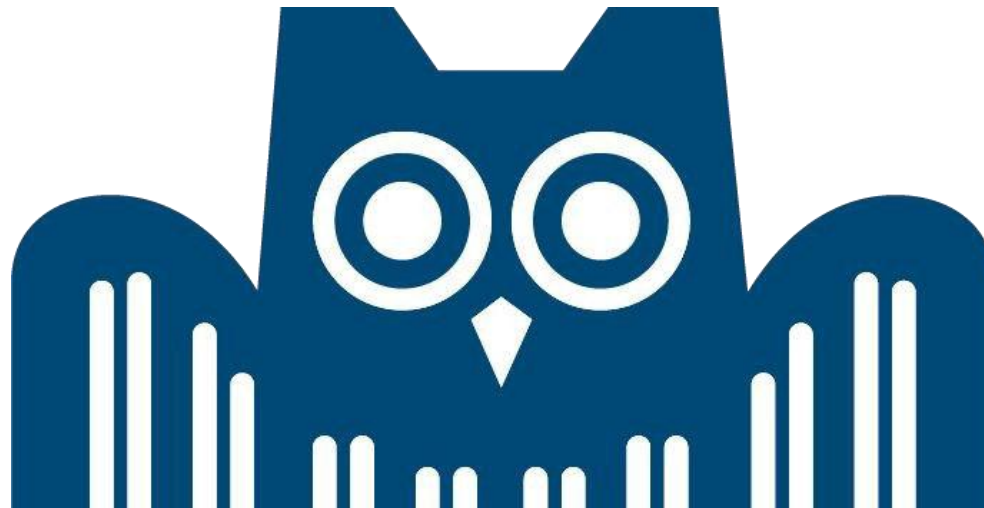
$\mathbf{A}_{i,j}$: Spatial adjacency between cell i and j

$\mathbf{z}_i, \mathbf{z}_j$: embedding of cell i and j

σ : sigmoid function

- 1** All tools were applied on the same Spatial Transcriptomics data.
- 2** Consistent input data and annotations were used across methods.
- 3** Method-specific internal preprocessing was applied as required.
- 4** Results were compared using spatial and interaction-based analyses.

Results



- Cell-type annotation provided directly.
- Single cell resolution.



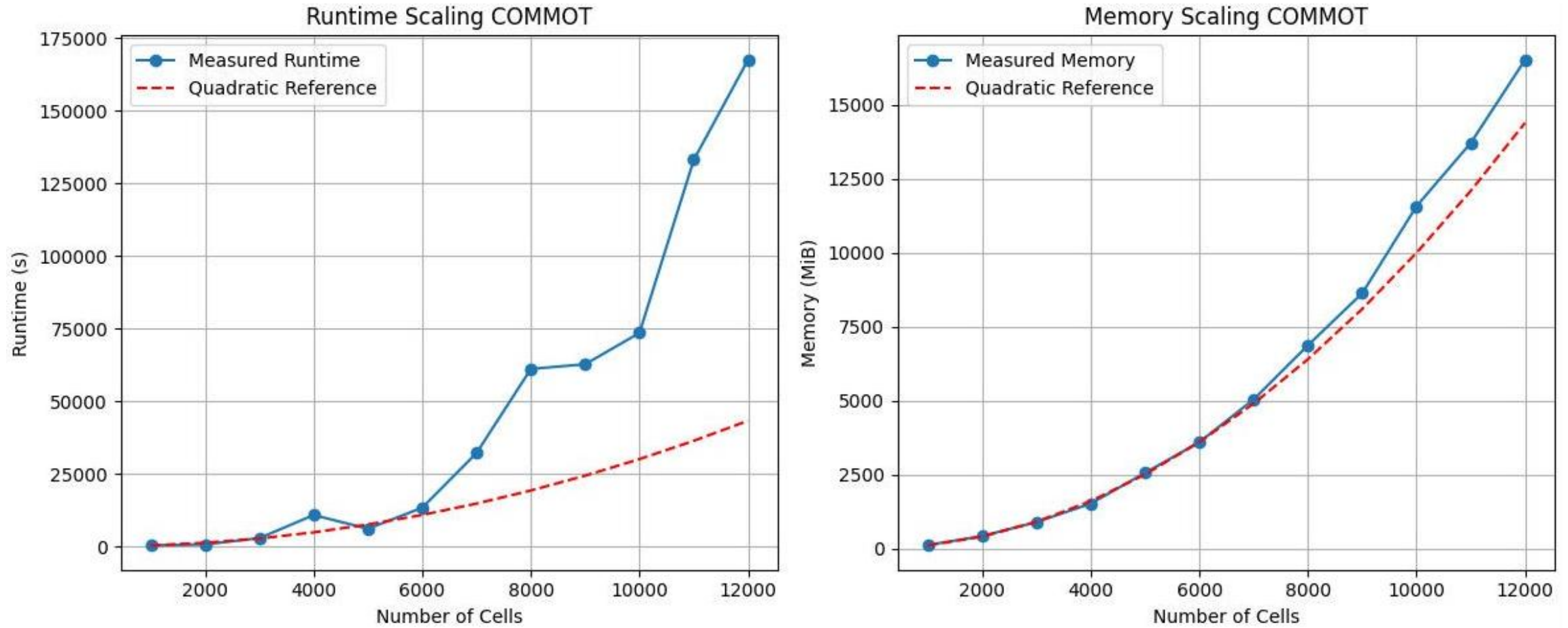


Figure 6: Scaling analysis for COMMOT

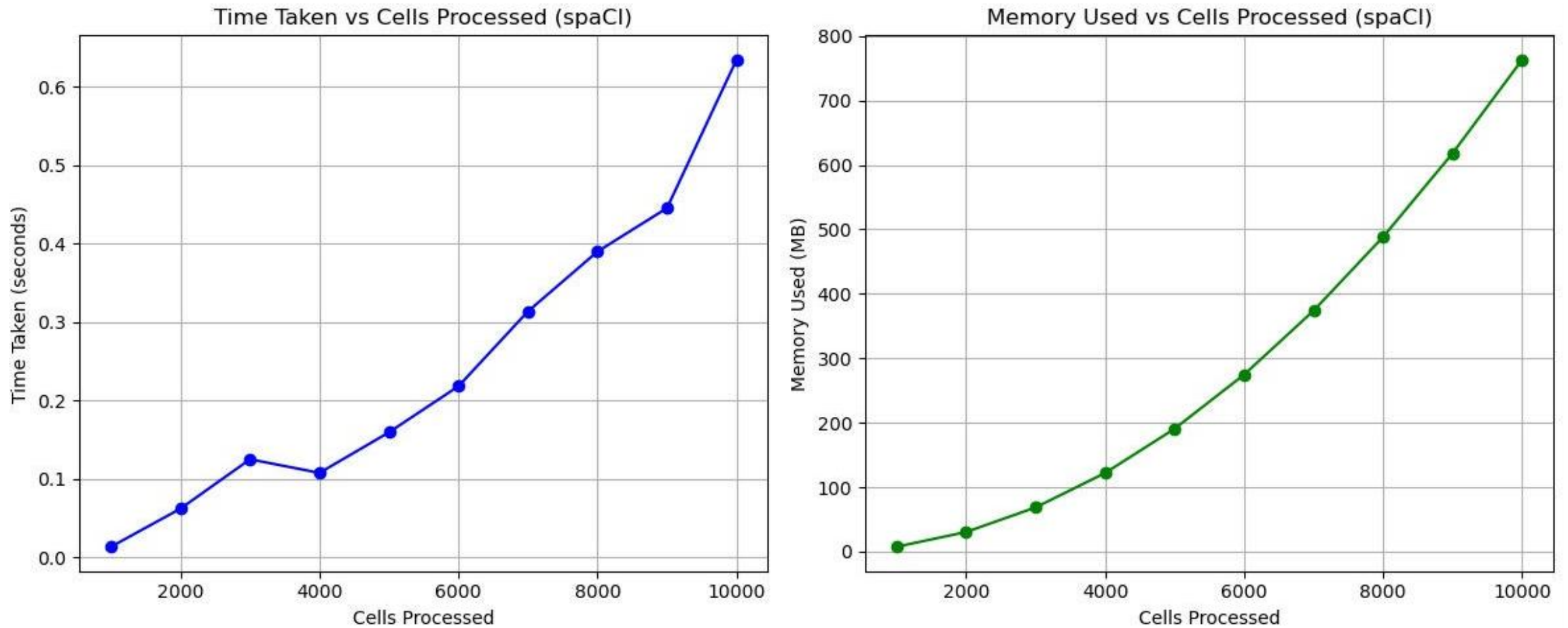


Figure 7: Scaling analysis for spaCI

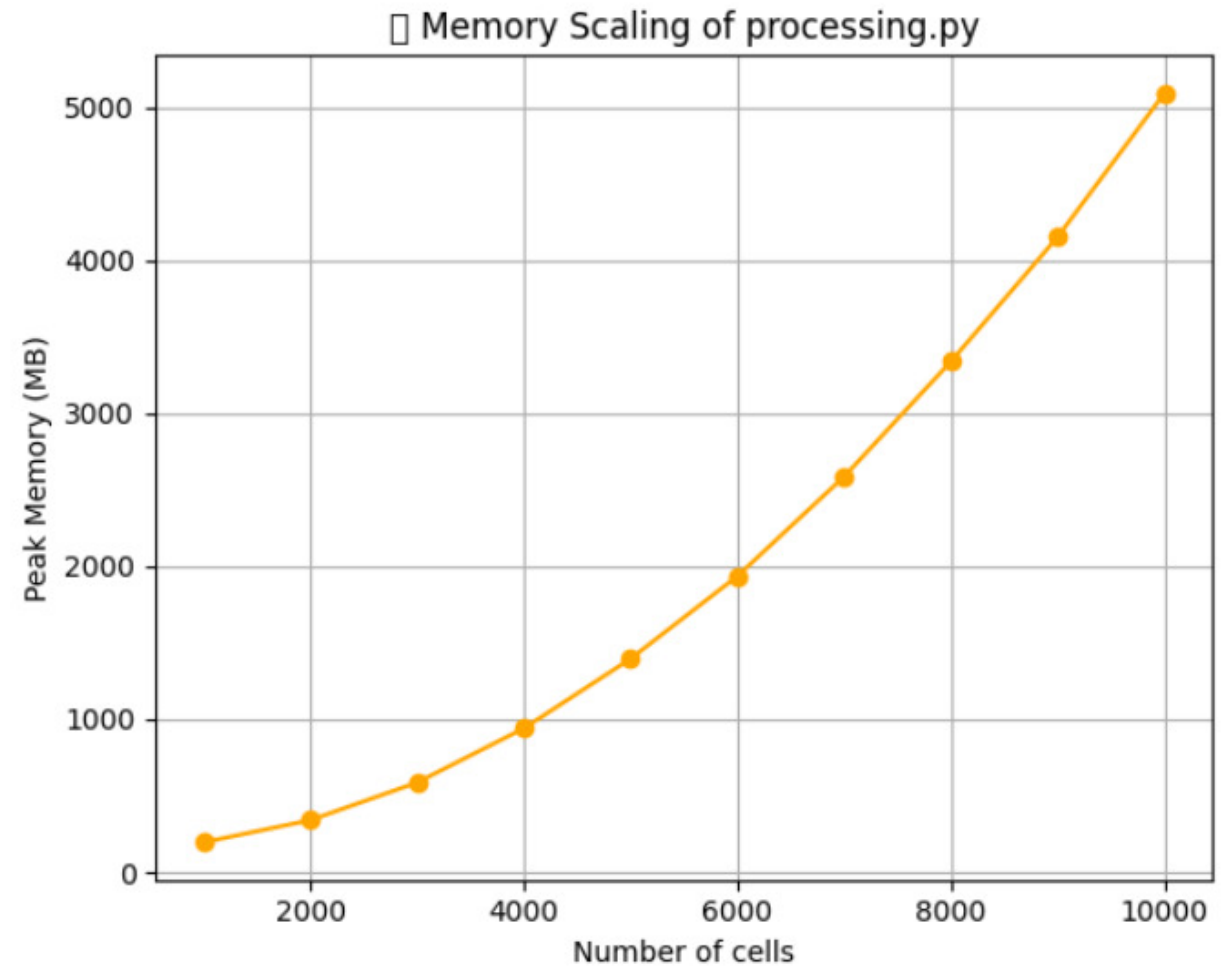
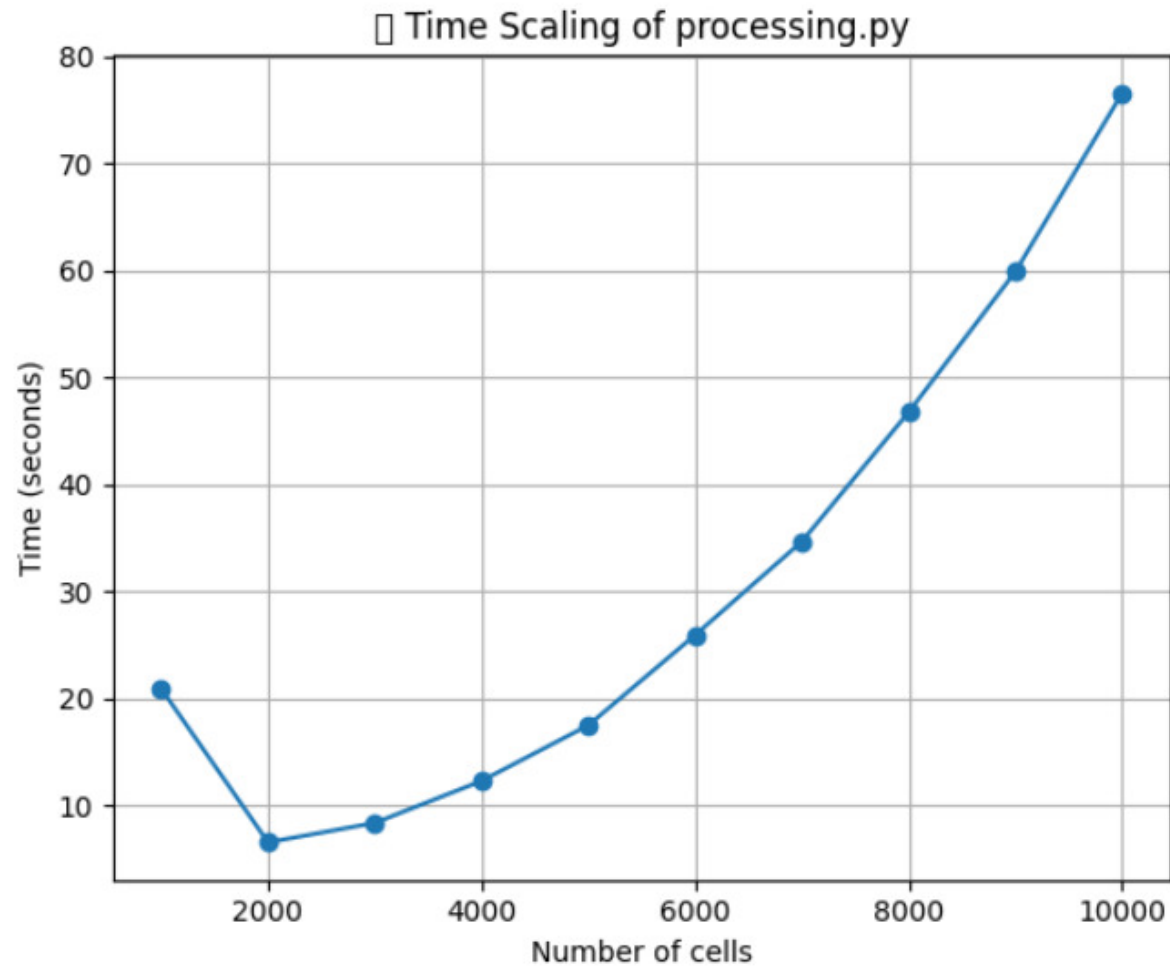
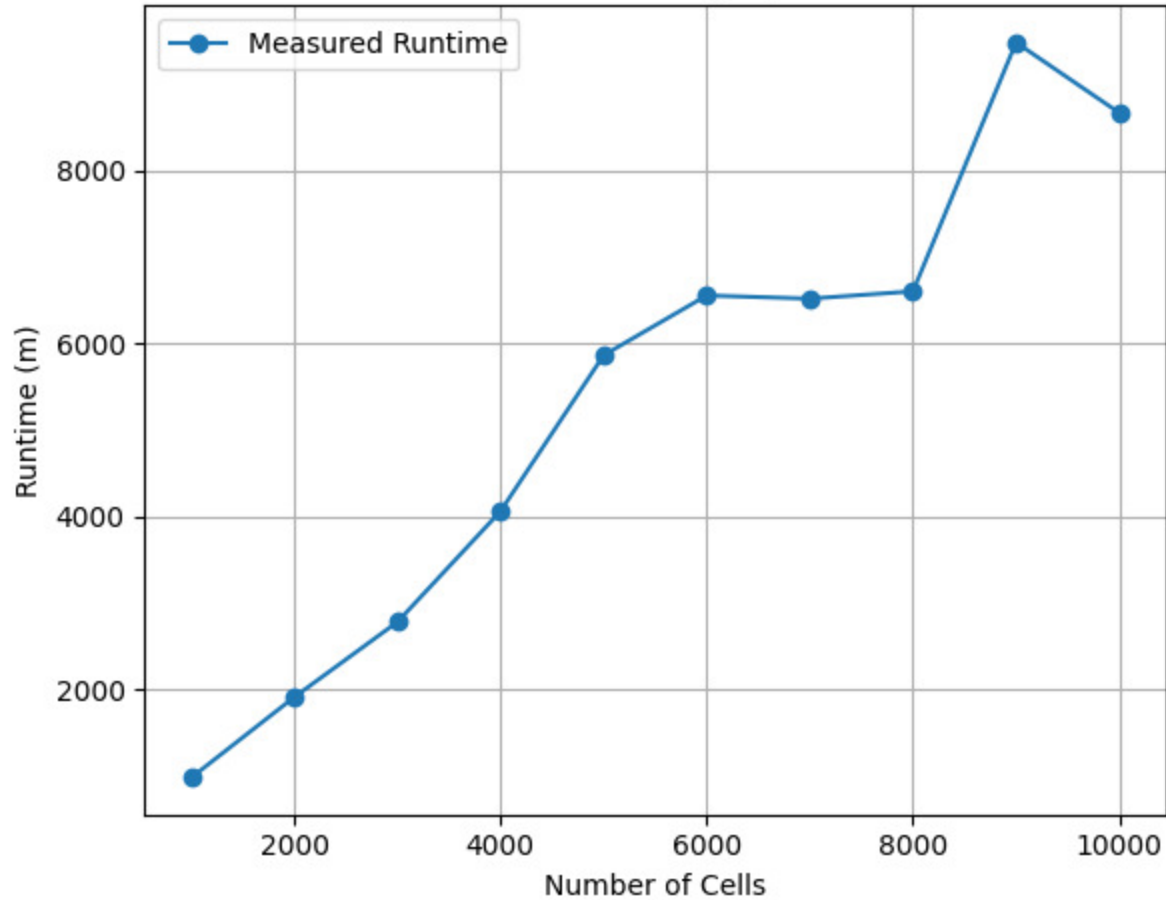


Figure 8: Scaling analysis for VGAE-CCI

Runtime Scaling Spacia



Memory Scaling Spacia

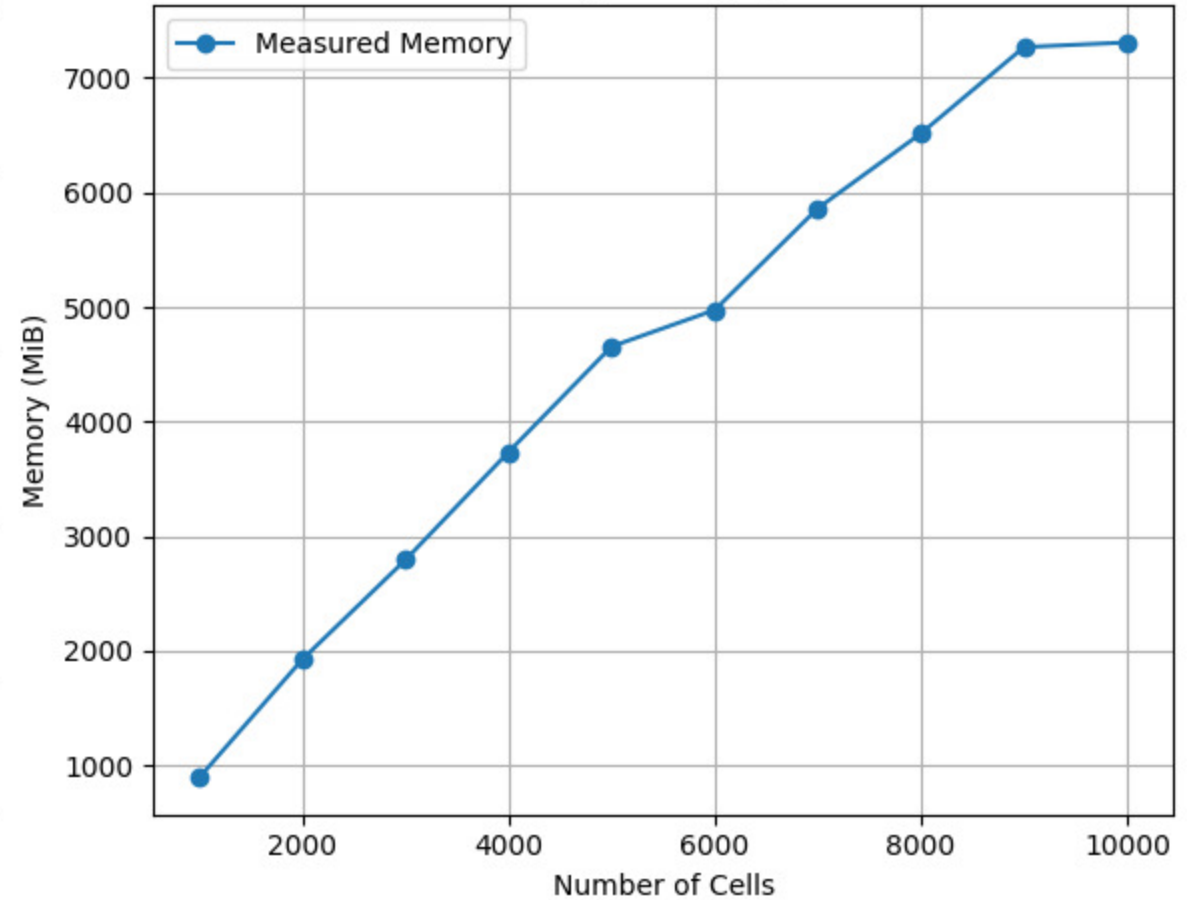


Figure 9: Scaling analysis for Spacia

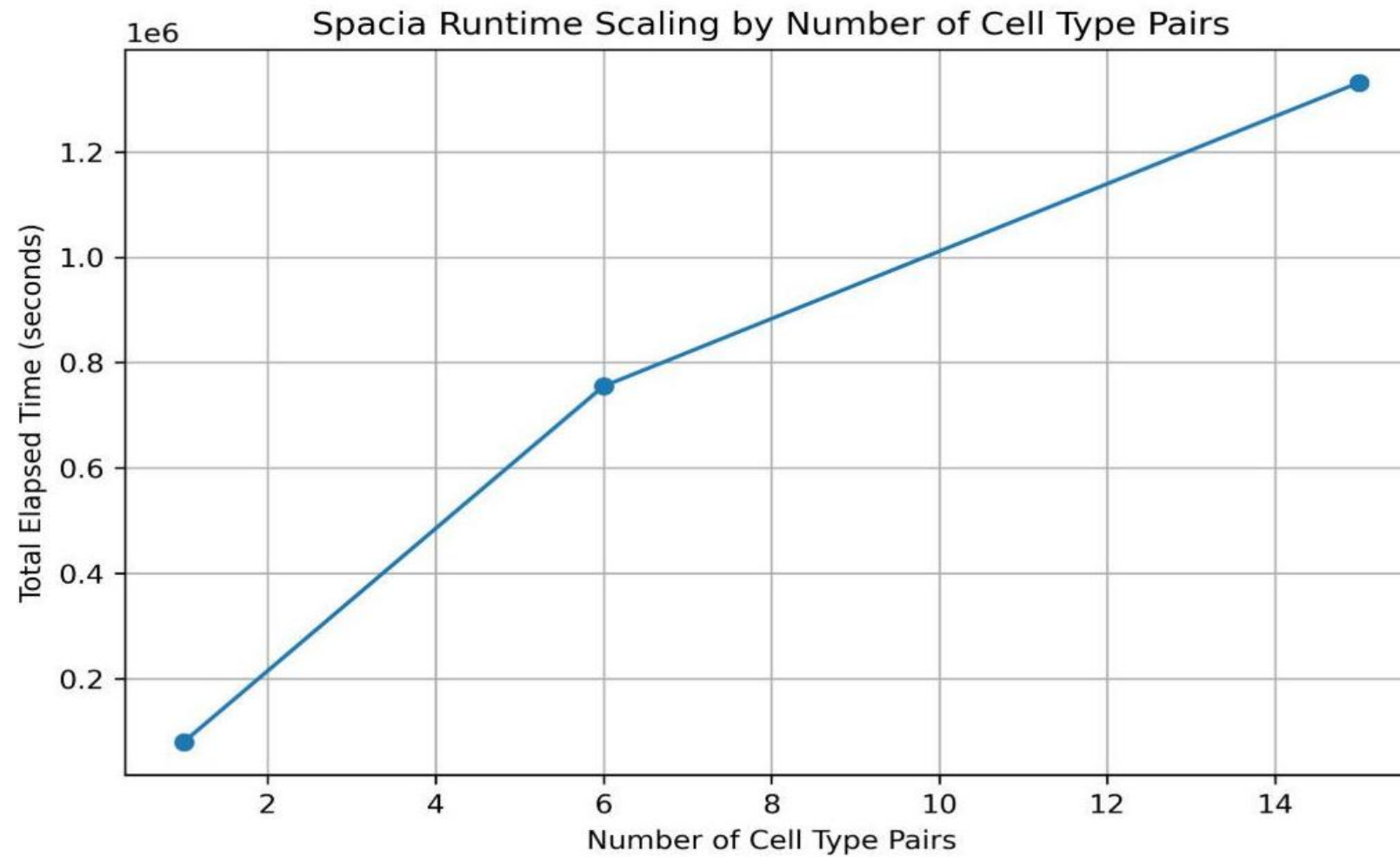


Figure 10: Scaling analysis for Spacia with increase number of pairs



Key Finding: All tools show quadratic behaviour and cannot run on the full 1,4M cells without subsetting except spacia.

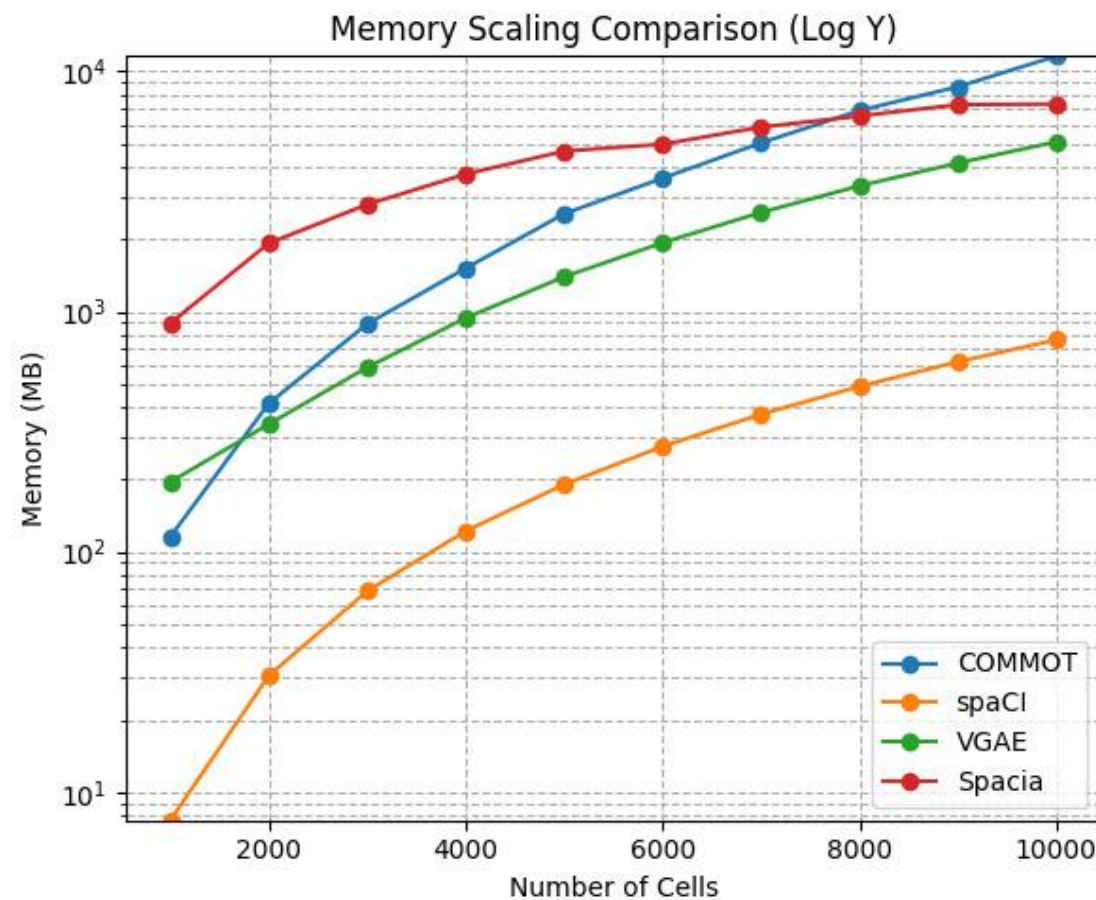
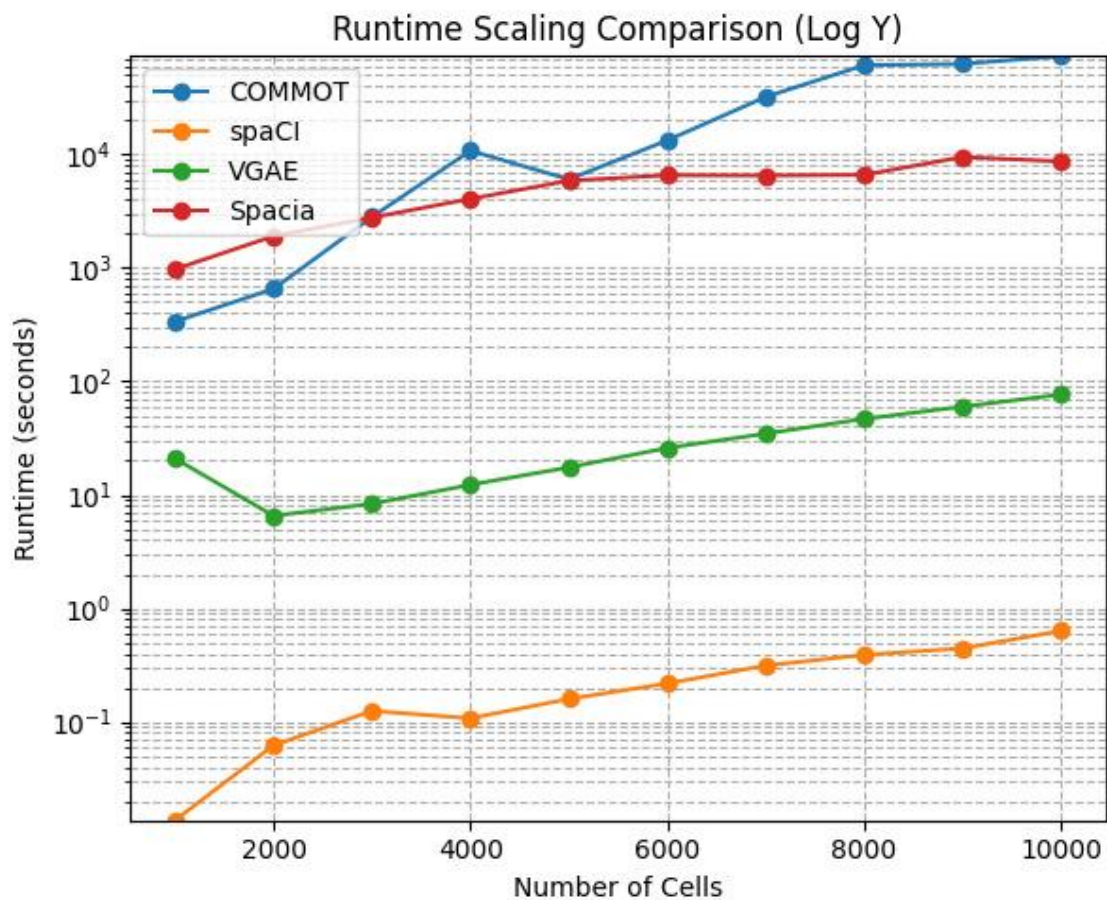
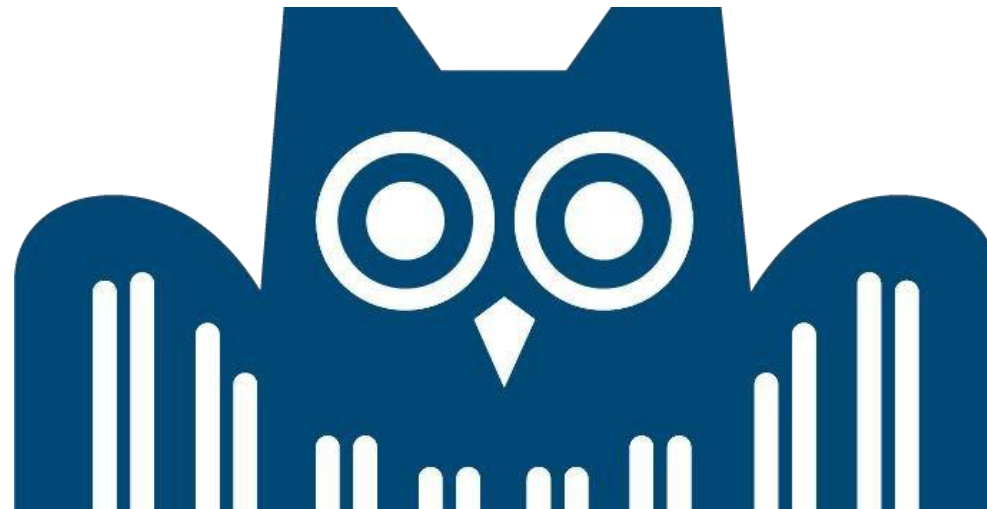


Figure 11: Scaling analysis for all tools with time log scale

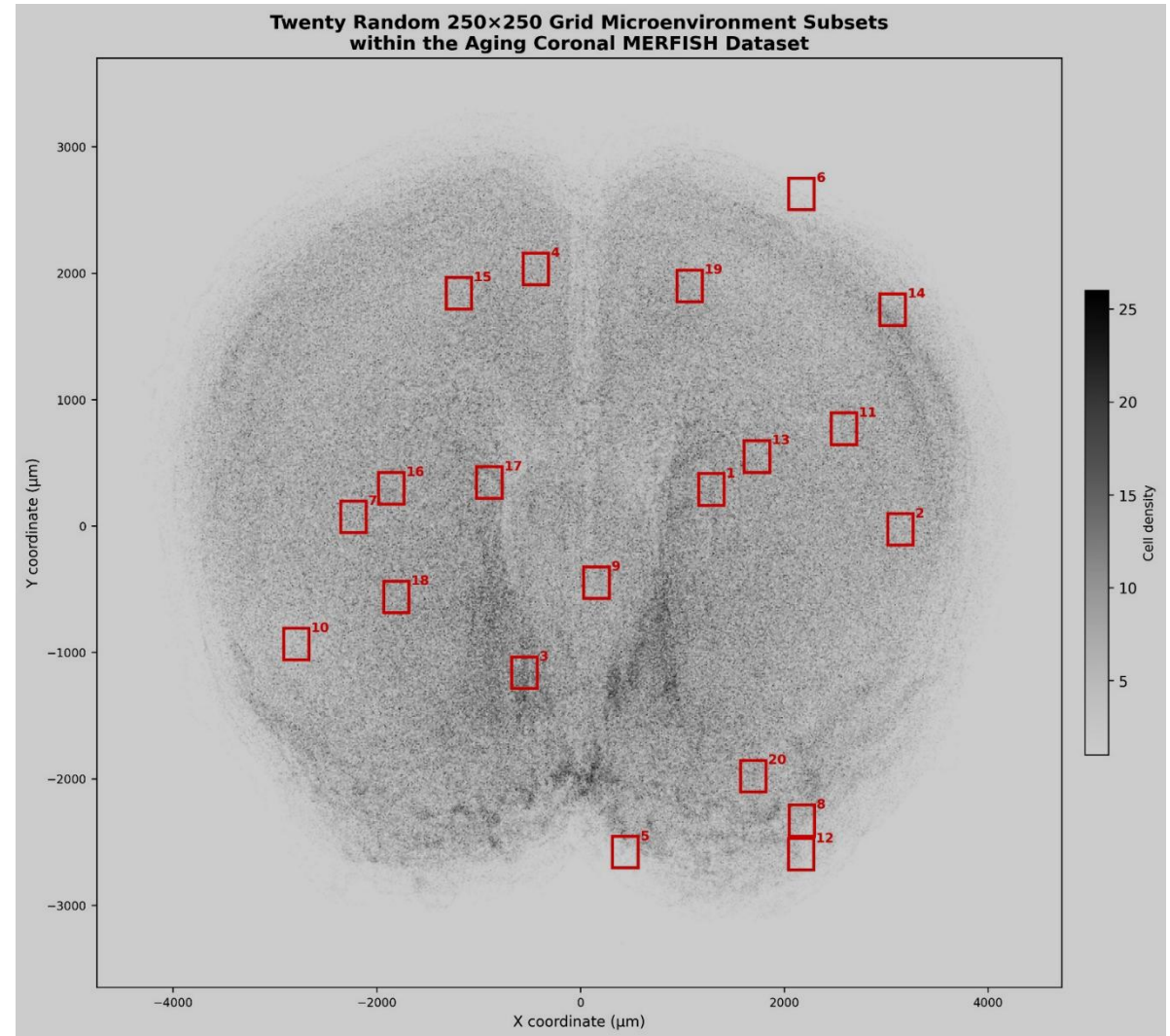
Microenvironment of Aging coronal



Coronal sample microenvironment

250 × 250 Grid-Based Microenvironment:

- Random anchor cell selected, and all cells within this window were extracted based on x-y coordinate.
- This yielded range between 500 - 3000 cells in each grid.



COMMOT result of Microenvironment

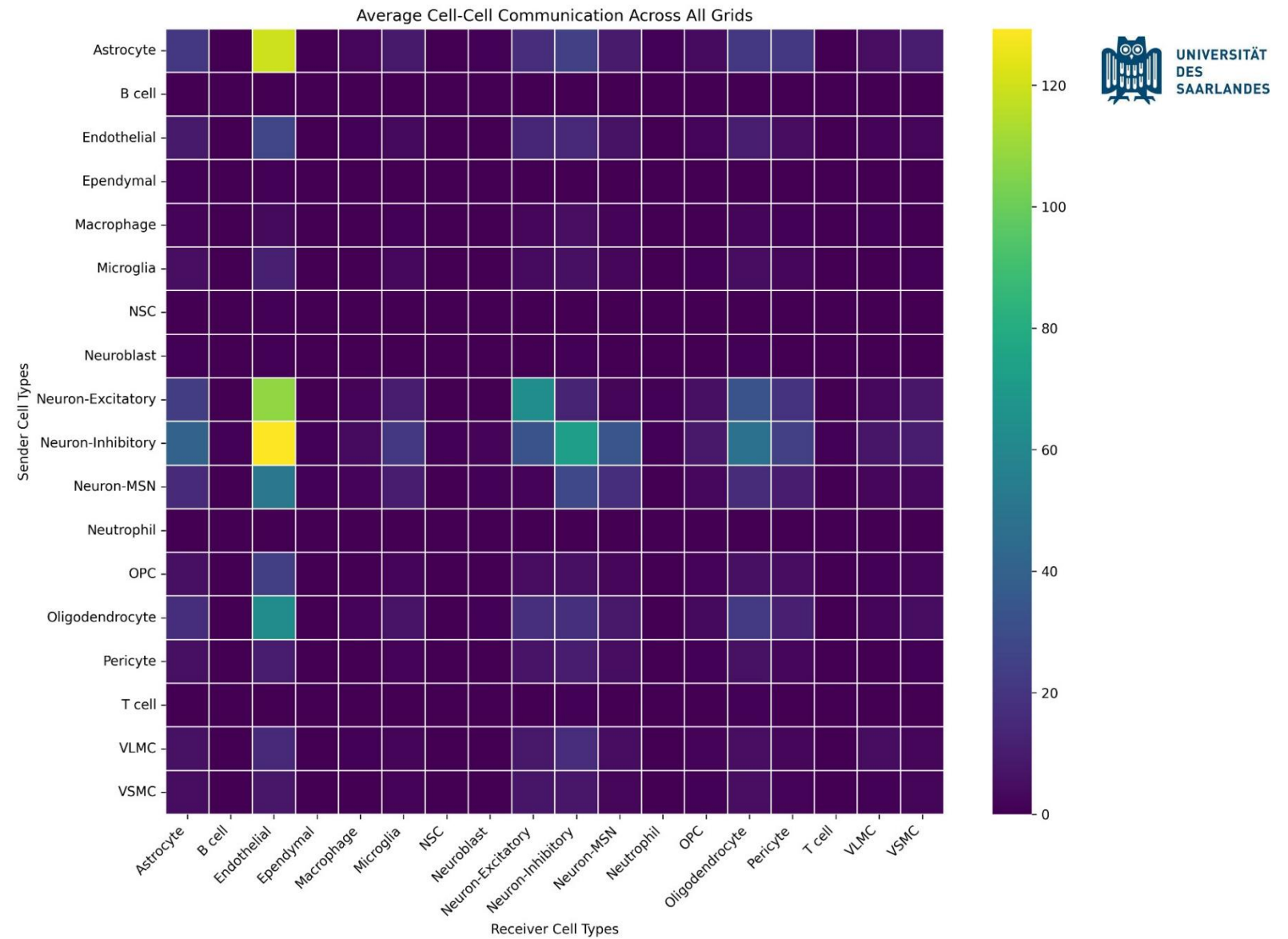


Figure 12: Signaling intensity result of COMMOT on microenvironments

spaCI result of Microenvironment

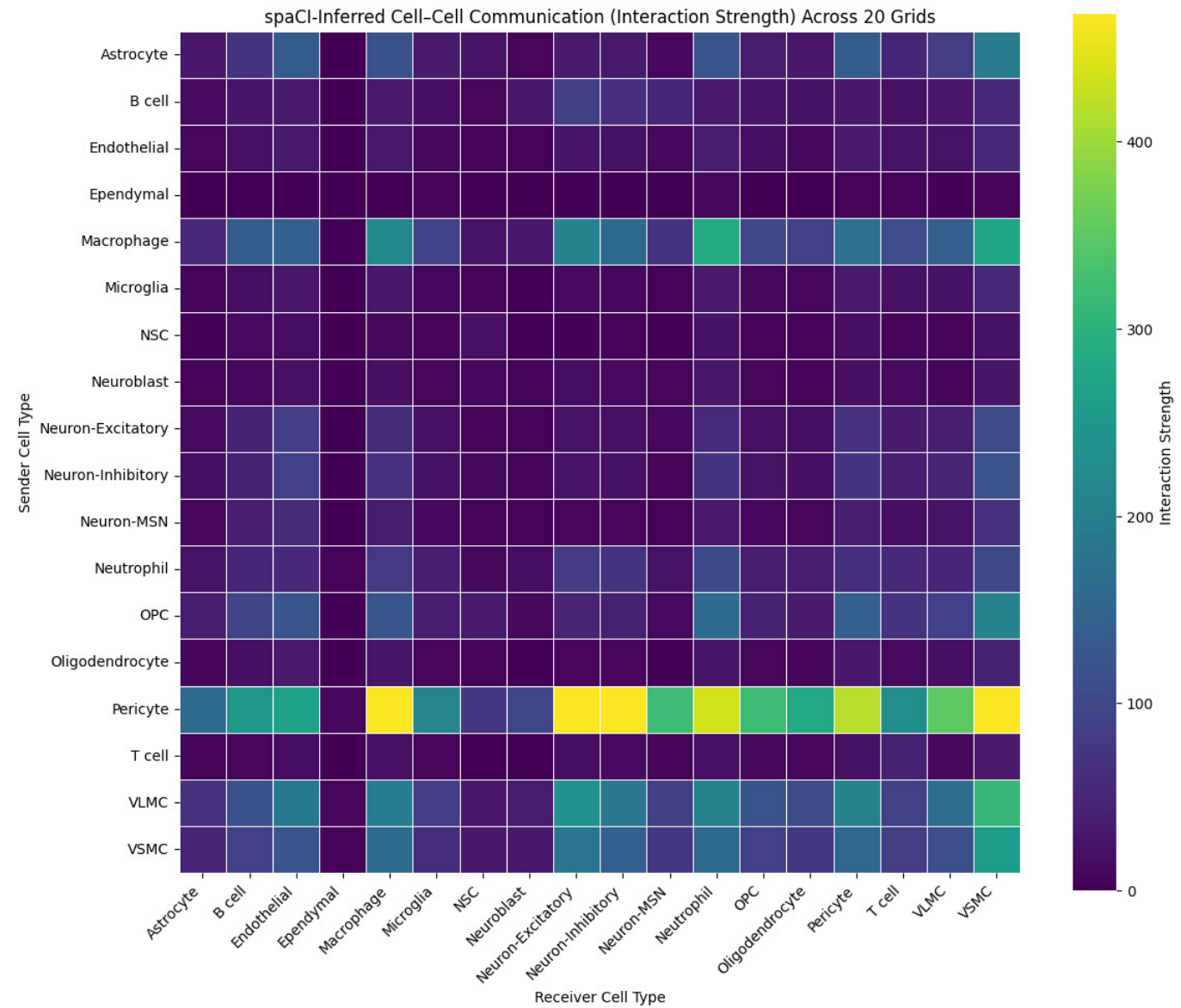


Figure 13: Interaction strength result of spaCI on microenvironments.

Spacia result of Microenvironment

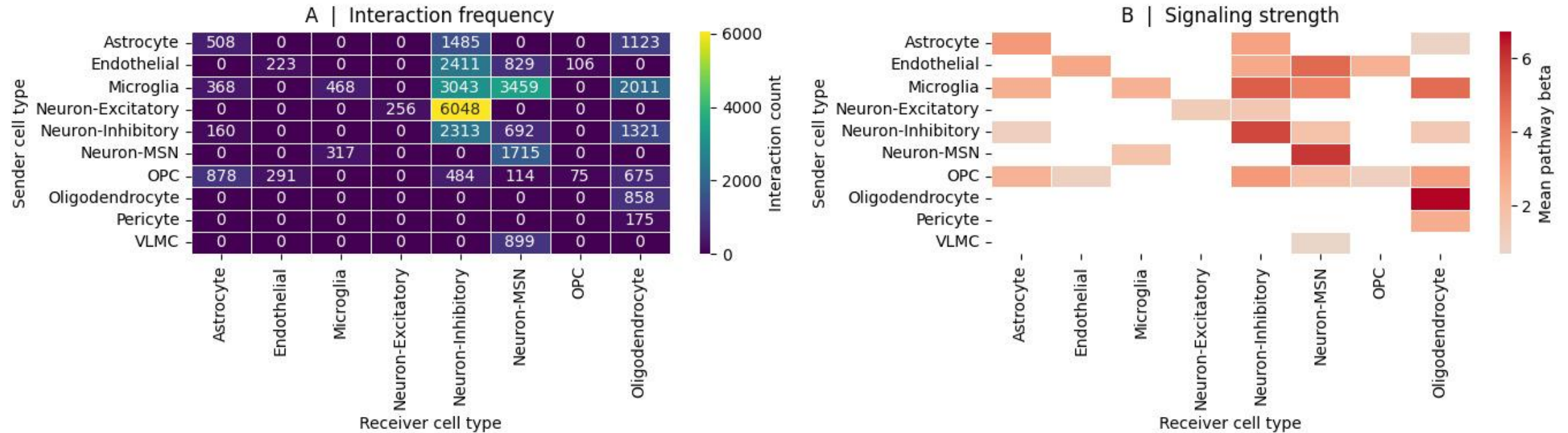


Figure 14: Spacia result of microenvironments, left image show the frequency between the cell types and the right one show the strength of interactions between the cell types.

VGAE result of Microenvironment

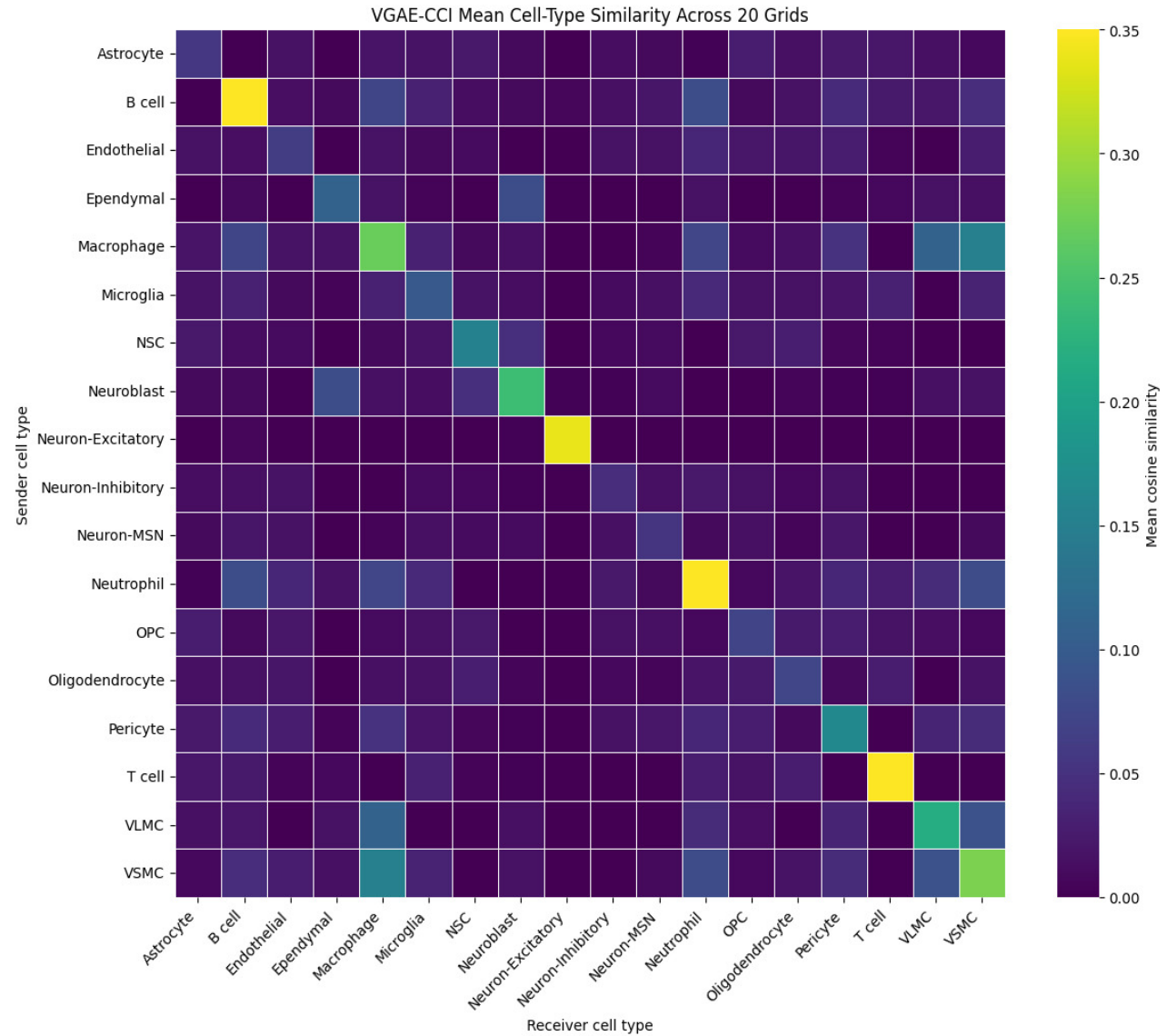


Figure 15: Cell type similarity heatmap based on VGAE embedding

Interaction pattern of Microenvironment

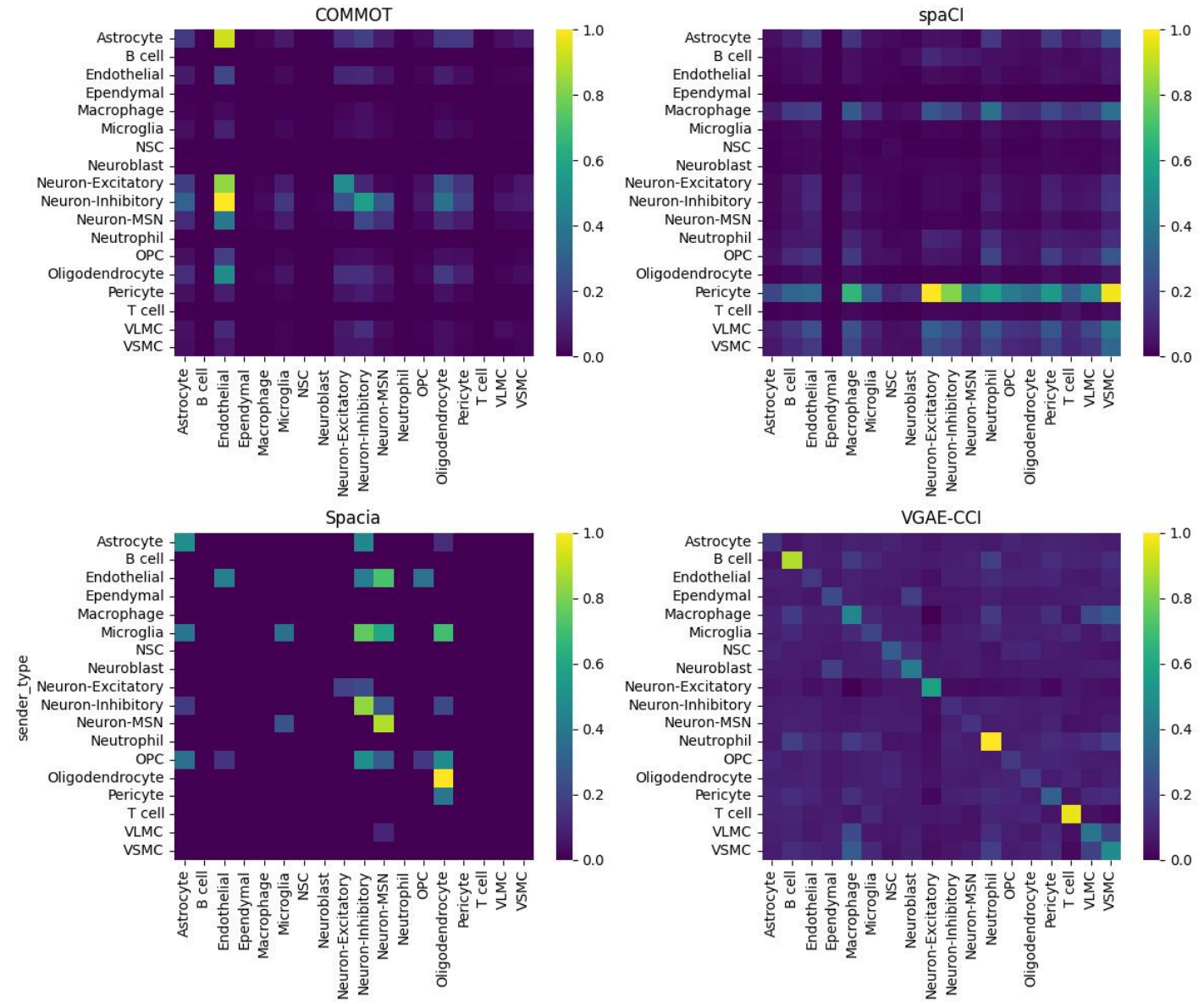


Figure 16: Overall interaction pattern for each tool, show strong and weak regions, by normalize each matrix result

Jaccard similarity between tools of **Microenvironment**

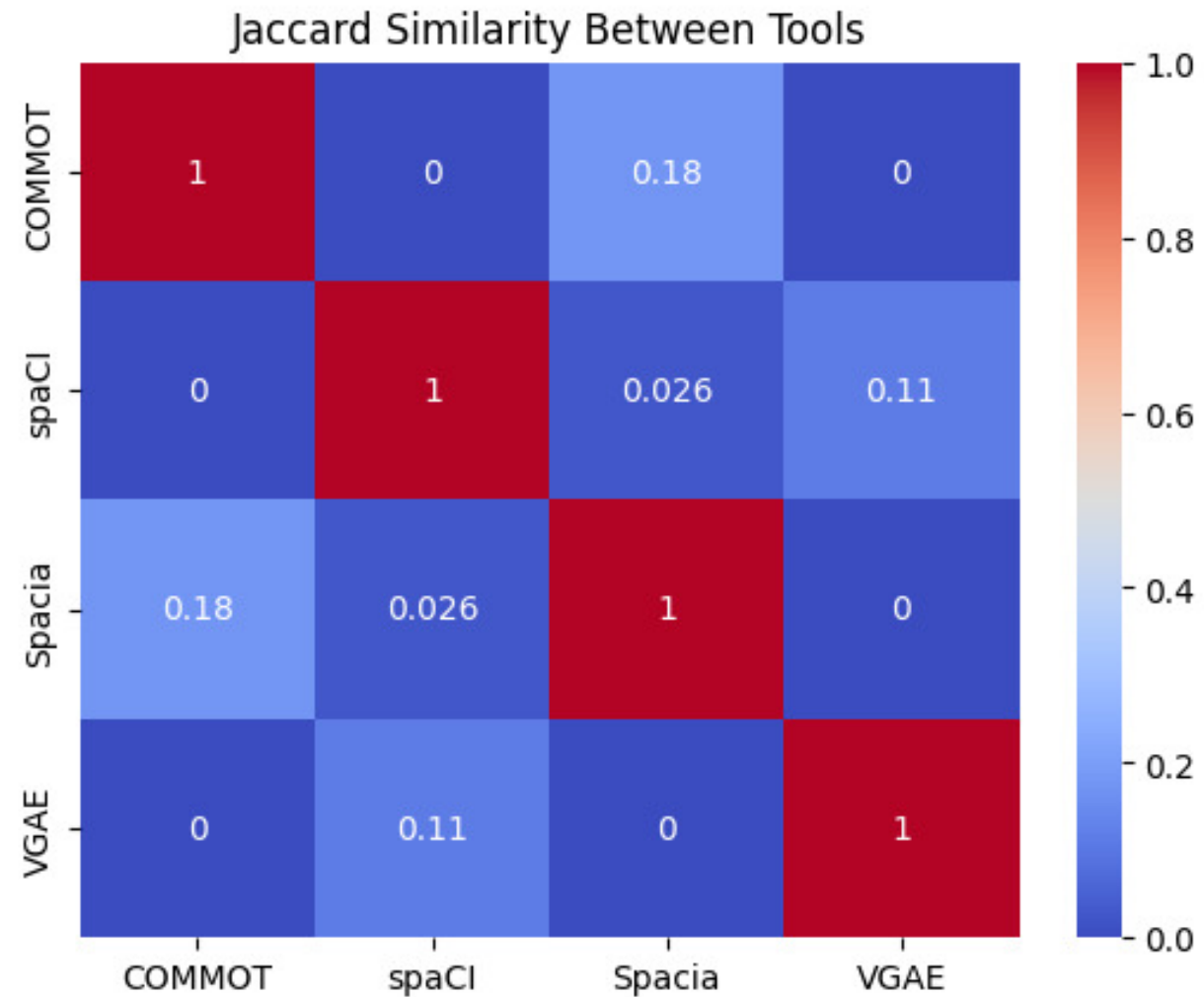
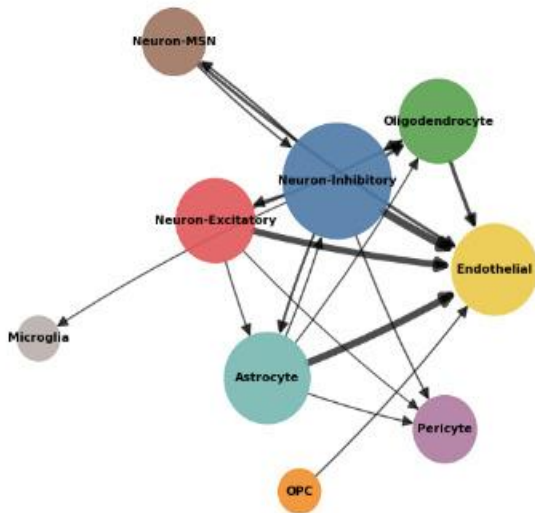


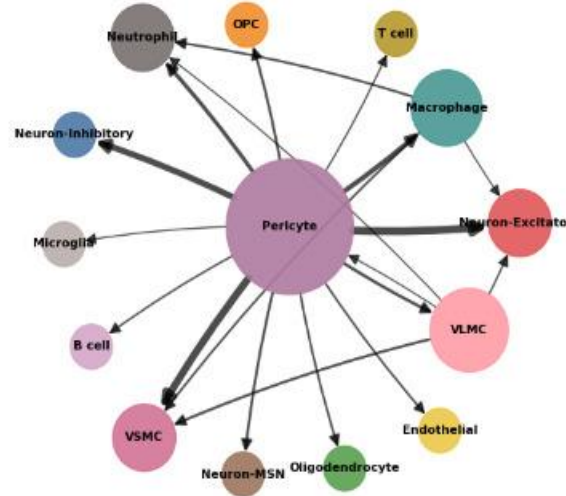
Figure 17: Jaccard – similarity interaction heatmap of microenvironment

Microenvironment Network

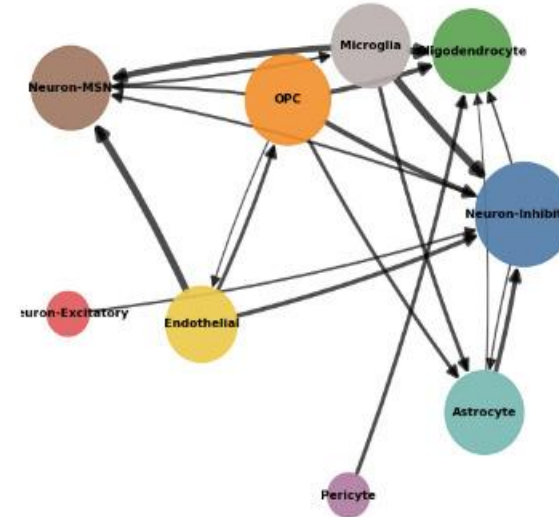
COMMOT Network



spaCI Network



Spacia Network



VGAE Network

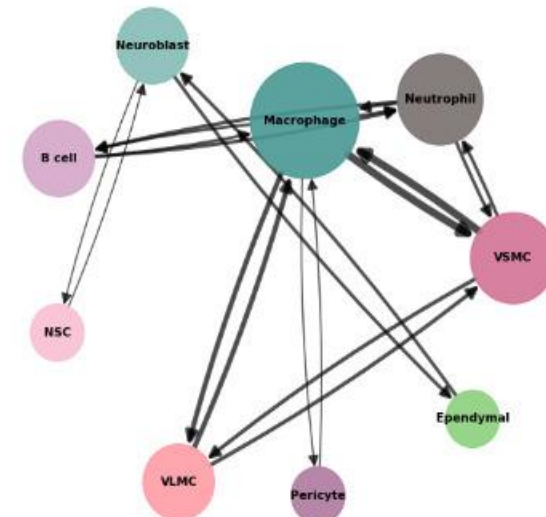
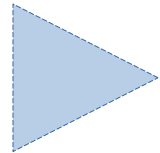


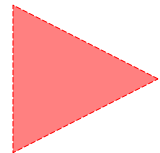
Figure 18: Network graph of microenvironment

Lower-Level Output



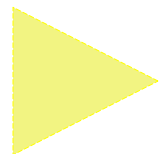
spaCI

- 44 L-R predict.
- 8 intersection.
- 36 unique.



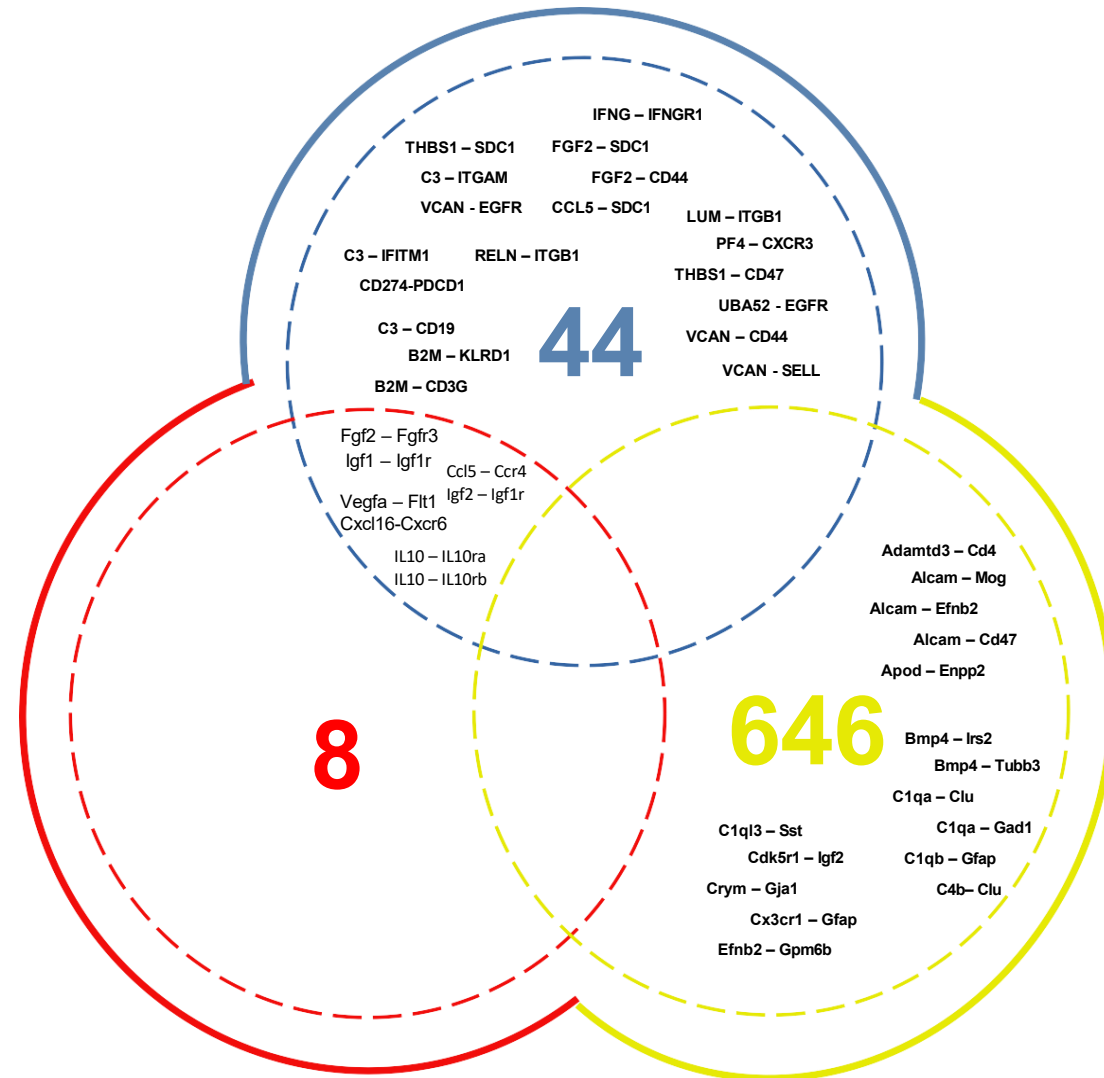
COMMOT

- 8 L-R predict.
- 8 intersection.
- None unique.

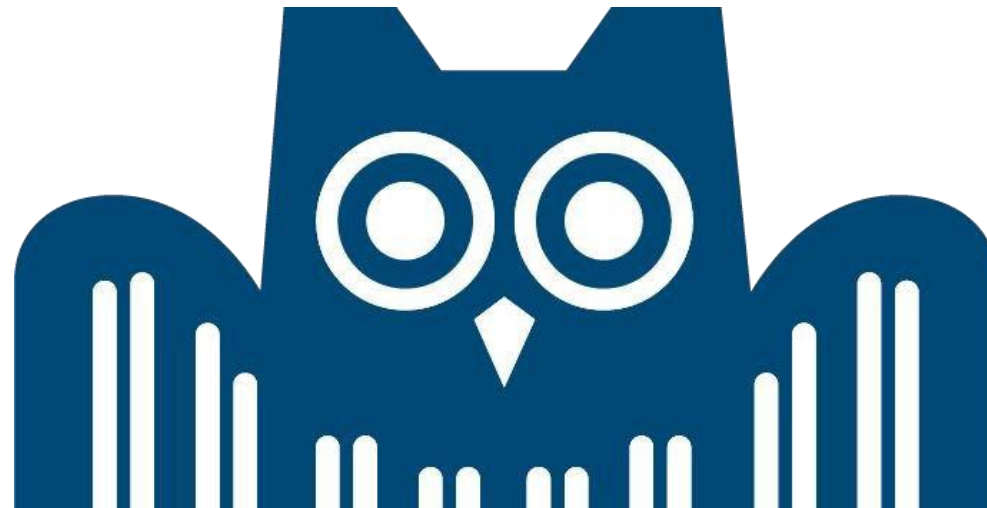


Spacia

- 646 gene-gene pair.



S1R1 sample



Vizgen sample (S1R1)

- S1R1 spatial dataset did not include the cell-type label.
- Did the single cell based annotation and assigned cell type.
- Add the cell type to the Meta file of the sample.



Annotation

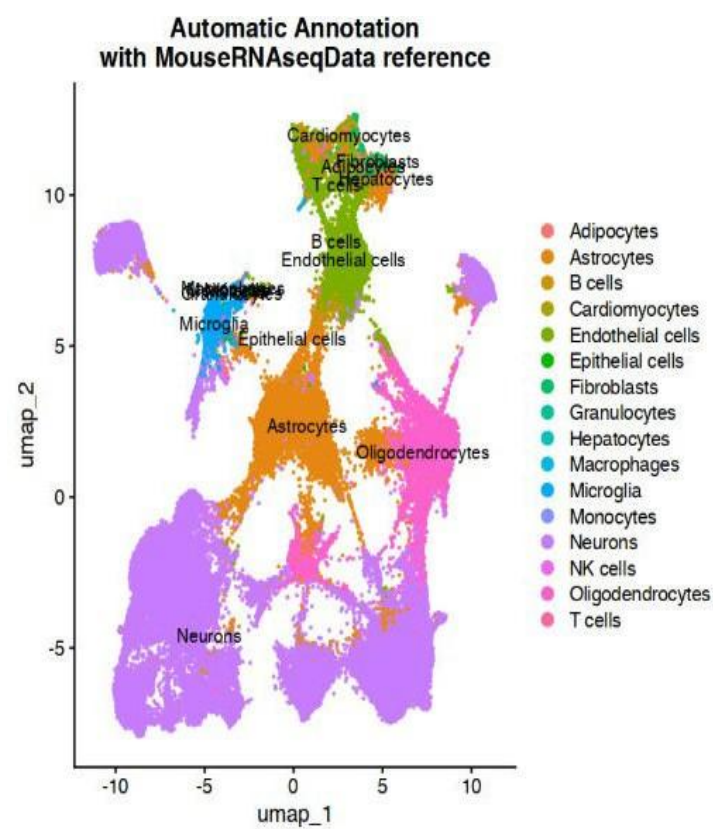


Figure 19: Automatic annotation of S1R1

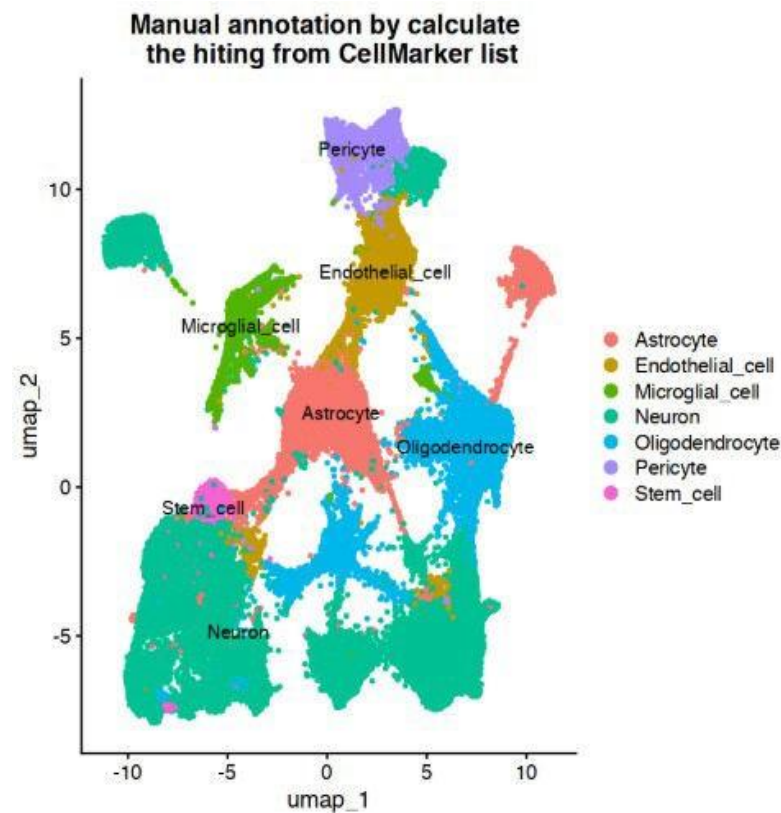


Figure 20: Manual annotation using CellMarker of S1R1

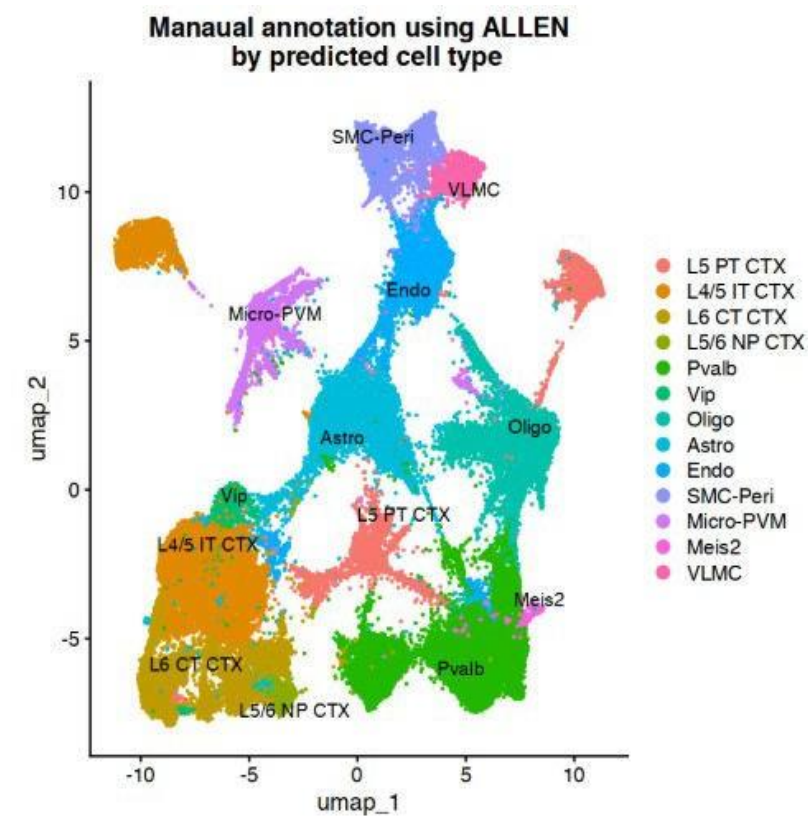


Figure 21: Manual annotation using ALLEN brain map of S1R1

Interaction pattern of S1R1

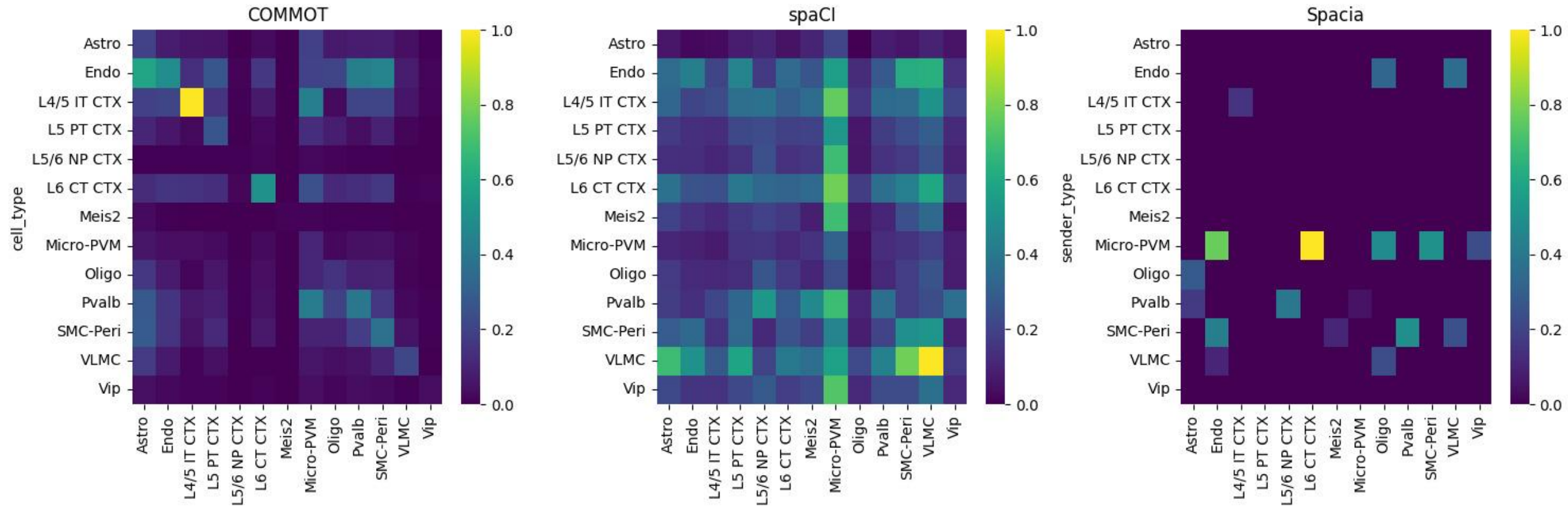
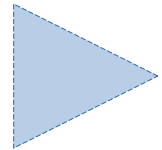


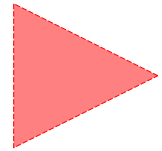
Figure 22: Overall interaction pattern for each tool, show strong and weak regions, by normalize each matrix result

Lower-Level Output



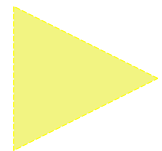
spaCI

- 19 L-R predict.
- 3 intersection.
- 16 unique.



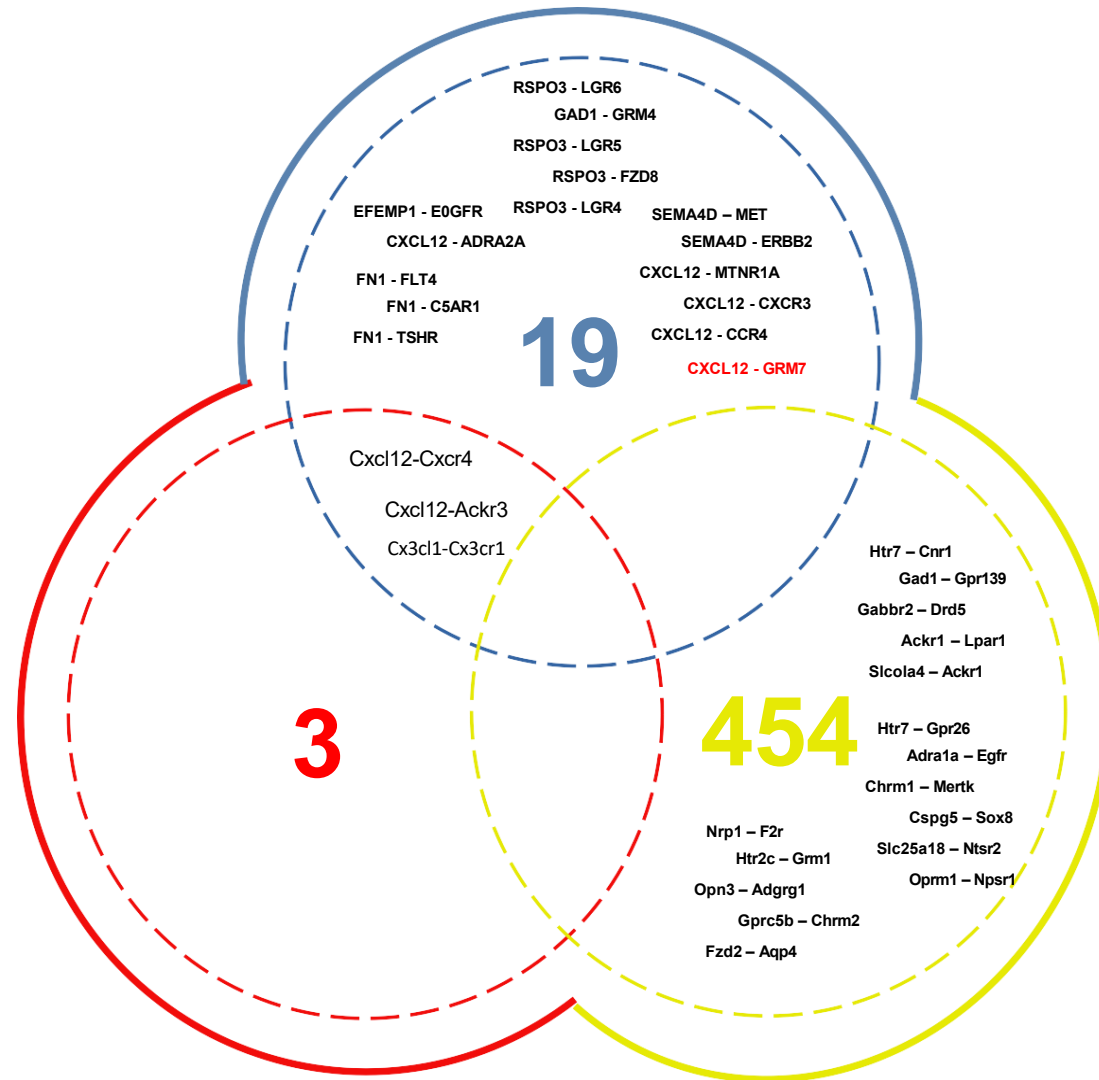
COMMOT

- 3 L-R predict.
- 3 intersection.
- None unique.

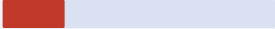
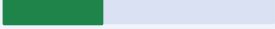


Spacia

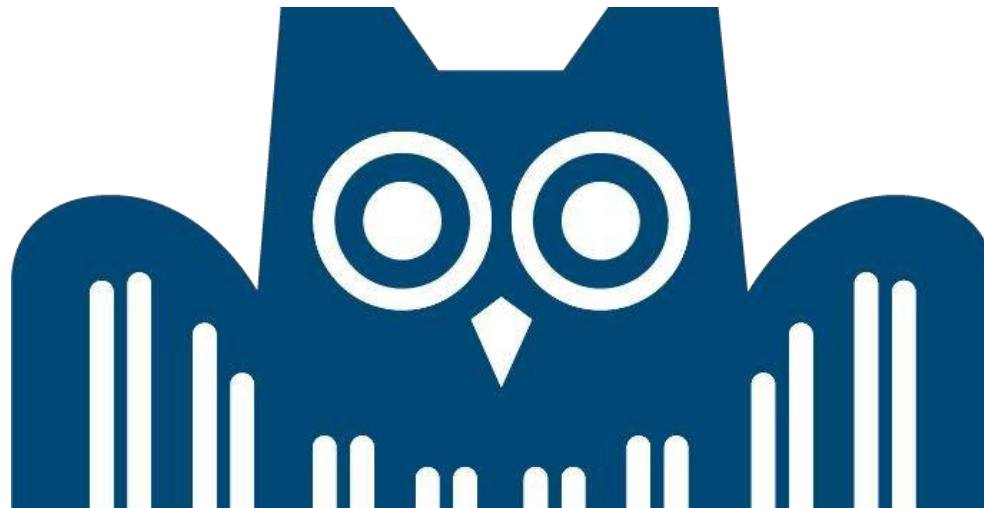
- 454 gene-gene pairs.



Comparing CCC Tools

Tool	Approach	Scalability	Documentation	Runtime	Prior Knowledge	Output Type
COMMOT	Optimal Transport (statistical)	 Poor	★★★★☆ Good	Hours / Days	Yes (CellChat)	L-R pairs
spaCI	Graph Neural Network (GNN)	 Poor	★★★★☆ Good	Hours / Days	Yes (own internal DB)	L-R Pair
Spacia	Bayesian MIL + MCMC	 Good	★★★★☆ Very Good	Hours / Days	No (data-driven)	Gene-gene influences
VGAE-CCI	Variational Graph Autoencoder	 Poor	★☆☆☆☆ Very Limited	Limited by input	No (data-driven)	Cell-type similarity

Conclusions



Conclusion

- No tool could run on the full 1.4M cell dataset without subsetting.
- Spacia handled larger subsets than others, but runtime increases with cell-type pair complexity.
- spaCI and COMMOT produced biologically interpretable results on moderate-sized data.
- VGAE-CCI captures cell-type similarity only — not suitable for L-R level inference.
- Zero overlap between Spacia and L-R tools reflects a difference in output type, not a biological contradiction.

- Provide a practical selection guide of CCC methods depending on dataset size and research goals.
- Highlights key trade-offs between scalability and interpretability.
- Demonstrates that no single tool is sufficient — combining methods gives a more complete CCC picture



Future work

Extend to more
datasets



Filter Spacia
output against
L-R DB for
comparison



Hybrid
statistical + ML
approaches
Rust
performance
gains



Address cell
morphology &
local
interaction bias



Thank You

